

4110_Prelim2_Group11

Sore Ajagbe, Alannah Dennis, Celina Jang, Cristina Rojo, Kai Wang

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The main objective of this assignment:

You should analyze and determine which factors might be predictive of a song being a “long-stay” song which stayed in the Number One spot for four or more weeks. You should also evaluate and quantify how those factors increase or decrease the probability of being a long-stay song.

The minor objective of this assignment:

You should discuss one characteristic that has shown a large shift in predictive importance over time. This may include a comparison of the key factors in predicting long- staying songs for two different decades (one prior to 2000 and one after 2000).

Data Cleaning

```
library(tidyverse)

## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.1      v stringr   1.5.2
## v ggplot2    4.0.0      v tibble    3.3.0
## v lubridate  1.9.4      v tidyr     1.3.1
## v purrr      1.1.0
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors

data <- read.csv('Billboard Top Songs v3 UPDATED.csv')
num_var_list <- c('Overall.Rating', 'Disagreement', 'BPM', 'Energy', 'Danceability', 'Happiness',
                  'Acousticness', 'Length')

# Data Cleaning
## identify any rows with missing or NA values
data[!complete.cases(data), ]

##      Song      Artist      Date Decade Long.Stay Weeks.at.Number.One
## 691 Rock On Michael Damian 3-Jun-89 1980          0              1
## 835 Lately      Divine 28-Nov-98 1990          0              1
##      Overall.Rating Disagreement Artist.Structure Artist.Male Artist.White
## 691          2.33          0.67          1          1          1
```

```
## 835          5.33          1.33          0          0          g'
##      Artist.Black BPM Energy Danceability Happiness Acousticness Vocally.Based
## 691          0 NA      NA          NA          NA          NA          0
## 835          1 NA      NA          NA          NA          NA          0
##      Bass.Based Guitar.Based Piano.Based Saxophone Trumpet Length
## 691          0          1          0          0          0      202
## 835          0          1          0          0          0      268
##      Written.for.a.Film
## 691          0
## 835          0
```

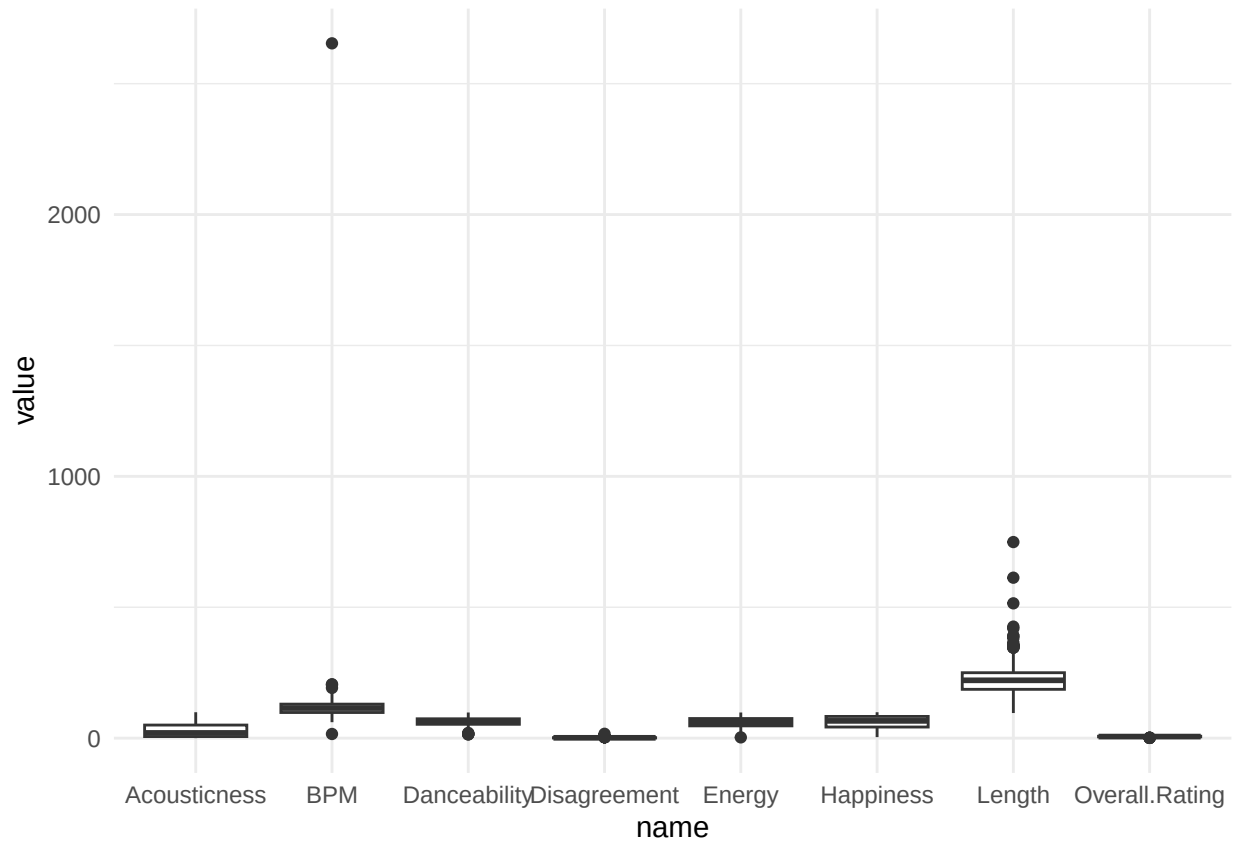
```
## remove rows with missing values and save as new data
data <- data[complete.cases(data), ]
```

```
## check any duplicates --> no duplicates found
data[duplicated(data), ]
```

```
## [1] Song          Artist          Date
## [4] Decade         Long.Stay      Weeks.at.Number.One
## [7] Overall.Rating Disagreement   Artist.Structure
## [10] Artist.Male    Artist.White   Artist.Black
## [13] BPM           Energy         Danceability
## [16] Happiness     Acousticness   Vocally.Based
## [19] Bass.Based    Guitar.Based   Piano.Based
## [22] Saxophone     Trumpet        Length
## [25] Written.for.a.Film
## <0 rows> (or 0-length row.names)
```

```
## check outliers
```

```
data |>
  pivot_longer(all_of(num_var_list)) |>
  ggplot(aes(x = name, y = value)) +
  geom_boxplot() +
  theme_minimal()
```



```
### Change codings
```

```
data <- data |>
  mutate(
    # Artist.Male has more than two categories.
    # The non-binary group only includes 5 observations, so we combine this with the mixed-gender
    # category.
    Artist.Sex = case_when(Artist.Male==0 ~ "All male",
                           Artist.Male==1 ~ "All female",
                           Artist.Male==2 | Artist.Male==3 ~ "Coed"),
    # Artist.Structure has no natural ordering
    Artist.Structure2 = case_when(Artist.Structure==0 ~ "3 or more",
                                   Artist.Structure==1 ~ "Solo",
                                   Artist.Structure==2 ~ "Duo"))
```

```
# - Artist.White has one observation with coding of 4, which is not explained in the codebook, so we remove it
data <- data[!(data$Artist.White=="4"),]
```

```
# - Additionally, we can combine the Artist.White and Artist.Black variables into one category, which would be
# one row has a group categorized as all white and all black. This is impossible, so we remove this observation
data <- data[!(data$Artist.White == "1" & data$Artist.Black == "1"),]
```

```
# recode
data <- data |>
```

```

mutate(Race = case_when(
  Artist.White == "1" ~ "All White",
  Artist.Black == "1" ~ "All Black",
  TRUE ~ "Mixed"
))

# - Guitar.Based has an observation with the coding 42, which is not explained in the codebook, so we r
data <- data[!(data$Guitar.Based == 42),]

# - BPM has an observation of 2654, which is an extreme outlier and is not the BPM of the song it is as
data$BPM[(data$BPM==2654)] <- 203
# - It also has an outlier of 16, associated with Against All Odds (Take a Look at Me Now) - Phil Colli
data$BPM[(data$BPM==16)] <- 116

# convert Artist.White column type
data <- data |> mutate(Artist.White = as.numeric(Artist.White))

# Rewrite updated dataset
write.csv(data, file = "cleaned_billboard.csv", row.names = FALSE)

```

Making Descriptive Tables

```

library(knitr)
library(kableExtra)

##
## Attaching package: 'kableExtra'

## The following object is masked from 'package:dplyr':
##
##   group_rows

# Distribution of variables
## categorical variables
cat_var_list <- c('Artist.Structure2', 'Artist.Sex', 'Race', 'Decade',
                  "Vocally.Based", "Bass.Based", "Guitar.Based", "Piano.Based", "Saxophone",
                  "Trumpet", "Written.for.a.Film")

label_map <- list(
  Long.Stay = c("0" = "Number One for 3 or fewer weeks",
                "1" = "Number One for 4+ weeks"),
  Vocally.Based = c("0" = "Not largely based", "1" = "Largely Based"),
  Bass.Based = c("0" = "Not largely based", "1" = "Largely Based"),
  Guitar.Based = c("0" = "Not largely based", "1" = "Largely Based"),
  Piano.Based = c("0" = "Not largely based", "1" = "Largely Based"),
  Saxophone = c("0" = "No", "1" = "Yes"),
  Trumpet = c("0" = "No", "1" = "Yes"),
  Written.for.a.Film = c("0" = "No", "1" = "Yes")
)

describe_cat <- function(var_list, outcome) {

```

```

dat <- data
n_total <- nrow(dat)
n_outcome0 <- sum(dat[[outcome]] == 0, na.rm = TRUE)
n_outcome1 <- sum(dat[[outcome]] == 1, na.rm = TRUE)

result <- tibble()

for (var_name in var_list) {
  dat2 <- dat

  # ensure character for grouping
  dat2[[var_name]] <- as.character(dat2[[var_name]])

  # apply label mapping to translate numerical coding
  if (var_name %in% names(label_map)) {
    dat2[[var_name]] <- sapply(dat2[[var_name]], function(x) label_map[[var_name]][[x]])
  }

  # overall counts and percents
  overall <- dat2 |>
    group_by(level = .data[[var_name]]) |>
    summarise(count_total = n(), .groups = "drop") |>
    mutate(percent_total = round(100 * count_total / n_total, 1))

  # counts within Long.Stay == 0
  outcome0 <- dat2 |>
    filter(.data[[outcome]] == 0) |>
    group_by(level = .data[[var_name]]) |>
    summarise(count_outcome0 = n(), .groups = "drop") |>
    mutate(percent_outcome0 = round(100 * count_outcome0 / n_outcome0, 1))

  # counts within Long.Stay == 1
  outcome1 <- dat2 |>
    filter(.data[[outcome]] == 1) |>
    group_by(level = .data[[var_name]]) |>
    summarise(count_outcome1 = n(), .groups = "drop") |>
    mutate(percent_outcome1 = round(100 * count_outcome1 / n_outcome1, 1))

  # join data sets
  joined <- overall |>
    full_join(outcome0, by = "level") |>
    full_join(outcome1, by = "level") |>
    replace_na(list(count_total = 0, percent_total = 0,
                    count_outcome0 = 0, percent_outcome0 = 0,
                    count_outcome1 = 0, percent_outcome1 = 0)) |>
    arrange(desc(count_total)) |>
    mutate(
      variable = var_name,
      `Overall (n, %)` = sprintf("%d (%.1f%%)", count_total, percent_total),
      `Long.Stay = 1 (n, %)` = sprintf("%d (%.1f%%)", count_outcome1, percent_outcome1)
    ) |>
    select(variable, level, `Overall (n, %)` , `Long.Stay = 1 (n, %)` )
}

```

```

# keep variable name only on first row for readability
if (nrow(joined) > 1) joined$variable[-1] <- ""

# Chi-square test for association with Long.Stay
tbl <- table(dat2[[var_name]], dat2[[outcome]])
chi_test <- chisq.test(tbl)
chi_stat <- round(chi_test$statistic, 2)
chi_p <- round(chi_test$p.value, 3)
df <- chi_test$parameter

# Add chi-square and p-value to the first row of each variable
joined$chisq <- ""
joined$chisq[1] <- paste0("X2 = ", chi_stat, ", df = ", df, ", p = ", chi_p)

result <- bind_rows(result, joined)
}

return(result)
}

cat_summary <- describe_cat(cat_var_list, "Long.Stay")

## Warning in chisq.test(tbl): Chi-squared approximation may be incorrect

# polish variable name for readability
cat_summary$variable <- gsub("\\.|[0-9]", " ", cat_summary$variable)
cat_summary$variable <- trimws(cat_summary$variable)

var_start_rows <- which(cat_summary$variable != "")
line_before <- var_start_rows[var_start_rows > 1] - 1

kable(cat_summary,

```

```
caption = "Descriptive Statistics of Categorical Variables by Long.Stay",
digits = 1,
col.names = c("Variable", "Level", 'Overall (n, %)',
              'Long.Stay = 1 (n, %)', 'Chi-square (p-value)'),
escape=F) |>
kable_styling(bootstrap_options = c("striped"), full_width= F) |>
row_spec(line_before, hline_after = TRUE)
```

Table 1: Descriptive Statistics of Categorical Variables by Long.Stay

Variable	Level	Overall (n, %)	Long.Stay = 1 (n, %)	Chi-square (p-value)
Artist Structure	Solo	667 (56.9%)	197 (64.0%)	$X^2 = 9.91$, df = 2, p = 0.007
	3 or more	408 (34.8%)	85 (27.6%)	
	Duo	97 (8.3%)	26 (8.4%)	
Artist Sex	All female	717 (61.2%)	170 (55.2%)	$X^2 = 8.12$, df = 2, p = 0.017
	All male	284 (24.2%)	80 (26.0%)	
	Coed	171 (14.6%)	58 (18.8%)	
Race	All White	634 (54.1%)	141 (45.8%)	$X^2 = 12.54$, df = 2, p = 0.002
	All Black	394 (33.6%)	118 (38.3%)	
	Mixed	144 (12.3%)	49 (15.9%)	
Decade	1970	253 (21.6%)	35 (11.4%)	$X^2 = 94.44$, df = 7, p = 0
	1980	228 (19.5%)	37 (12.0%)	
	1960	202 (17.2%)	44 (14.3%)	
	1990	139 (11.9%)	51 (16.6%)	
	2000	129 (11.0%)	56 (18.2%)	
	2010	116 (9.9%)	56 (18.2%)	
	2020	82 (7.0%)	19 (6.2%)	
	1950	23 (2.0%)	10 (3.2%)	
Vocally Based	Not largely based	1106 (94.4%)	293 (95.1%)	$X^2 = 0.28$, df = 1, p = 0.595
	Largely Based	66 (5.6%)	15 (4.9%)	
Bass Based	Not largely based	907 (77.4%)	232 (75.3%)	$X^2 = 0.86$, df = 1, p = 0.353
	Largely Based	265 (22.6%)	76 (24.7%)	
Guitar Based	Largely Based	655 (55.9%)	143 (46.4%)	$X^2 = 14.65$, df = 1, p = 0
	Not largely based	517 (44.1%)	165 (53.6%)	
Piano Based	Largely Based	592 (50.5%)	153 (49.7%)	$X^2 = 0.08$, df = 1, p = 0.783
	Not largely based	580 (49.5%)	155 (50.3%)	
Saxophone	No	1110 (94.7%)	295 (95.8%)	$X^2 = 0.69$, df = 1, p = 0.408
	Yes	62 (5.3%)	13 (4.2%)	
Trumpet	No	1153 (98.4%)	305 (99.0%)	$X^2 = 0.62$, df = 1, p = 0.433
	Yes	19 (1.6%)	3 (1.0%)	
Written for a Film	No	1080 (92.2%)	281 (91.2%)	$X^2 = 0.33$, df = 1, p = 0.567
	Yes	92 (7.8%)	27 (8.8%)	

```

## numerical variables
describe_num <- function(var_list, outcome) {
  results <- data.frame()

  for (var_name in var_list) {
    # likelihood ratio test
    model_full <- glm(as.formula(paste(outcome, "~", var_name)),
                      data = data, family = binomial)
    model_null <- glm(as.formula(paste(outcome, "~ 1")),
                     data = data, family = binomial)
    G2 <- deviance(model_null) - deviance(model_full)
    p_value <- round(pchisq(G2, df = 1, lower.tail = FALSE), 3)

    # summary stats by Long.Stay
    stat_table <- data |>
      summarise(
        mean = round(mean(data[[var_name]]), na.rm = TRUE), 2),
        median = round(median(data[[var_name]]), na.rm = TRUE), 2),
        sd = round(sd(data[[var_name]]), na.rm = TRUE), 2),
        min = round(min(data[[var_name]]), na.rm = TRUE), 2),
        max = round(max(data[[var_name]]), na.rm = TRUE), 2),
      ) |>
      mutate(
        `Mean (SD)` = sprintf("%.2f (%.2f)", mean, sd)
      )

    # adjust formatting
    wide_stats <- tibble(
      Variable = var_name,
      `Mean (SD)` = stat_table$`Mean (SD)`,
      Median = stat_table$median,
      Min = stat_table$min,
      Max = stat_table$max,
      `p-value` = p_value
    )

    results <- bind_rows(results, wide_stats)
  }

  return(results)
}

num_summary <- describe_num(num_var_list, "Long.Stay")

# polish variable name for readability
num_summary$Variable <- gsub("\\.|[0-9]", " ", num_summary$Variable)
num_summary$Variable <- trimws(num_summary$Variable)

# determine which rows should get a line before them
var_start_rows <- which(num_summary$variable != "")
line_before <- var_start_rows[var_start_rows > 1] - 1

```

```
kable(num_summary, caption = "Descriptive Statistics of Numerical Variables") |>
  kable_styling(bootstrap_options = c("striped")) |>
  row_spec(line_before, hline_after = TRUE)
```

Table 2: Descriptive Statistics of Numerical Variables

Variable	Mean (SD)	Median	Min	Max	p-value
Overall Rating	6.14 (1.82)	6.16	1	10.00	0.000
Disagreement	1.57 (1.06)	1.33	0	16.66	0.052
BPM	115.97 (24.48)	115.00	61	206.00	0.607
Energy	60.49 (19.27)	61.00	3	98.00	0.918
Danceability	62.97 (15.21)	64.00	14	98.00	0.002
Happiness	61.78 (24.79)	66.00	4	99.00	0.314
Acousticness	29.96 (27.49)	20.00	0	99.00	0.019
Length	221.74 (52.81)	221.00	96	749.00	0.040

Contingency Table

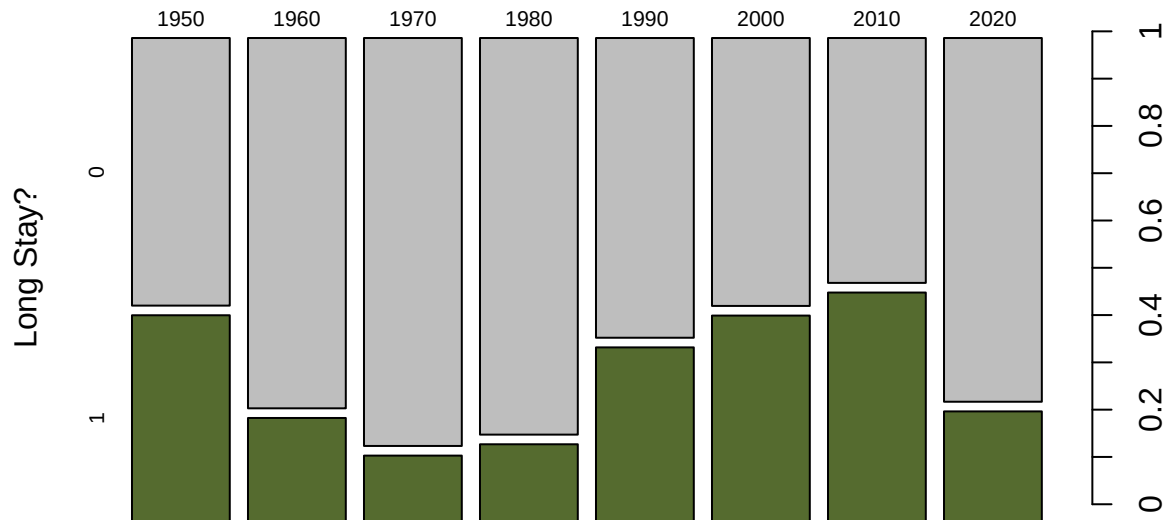
```
attach(data)

# We use proportions instead of pure counts to more easily visualize the relationships.

# a function to facilitate printing contingency tables
tprint <- function(varname) {
  tab <- prop.table(table(Long.Stay, get(varname)), 2)
  t(tab)
}

plot(tprint("Decade"), main="Decades vs Long-stay Rate", ylab="Long Stay?",
     col=c("gray", "darkolivegreen"))
axis(4, at = seq(0, 1, by = 0.1), labels = seq(0, 1, by = 0.1))
```

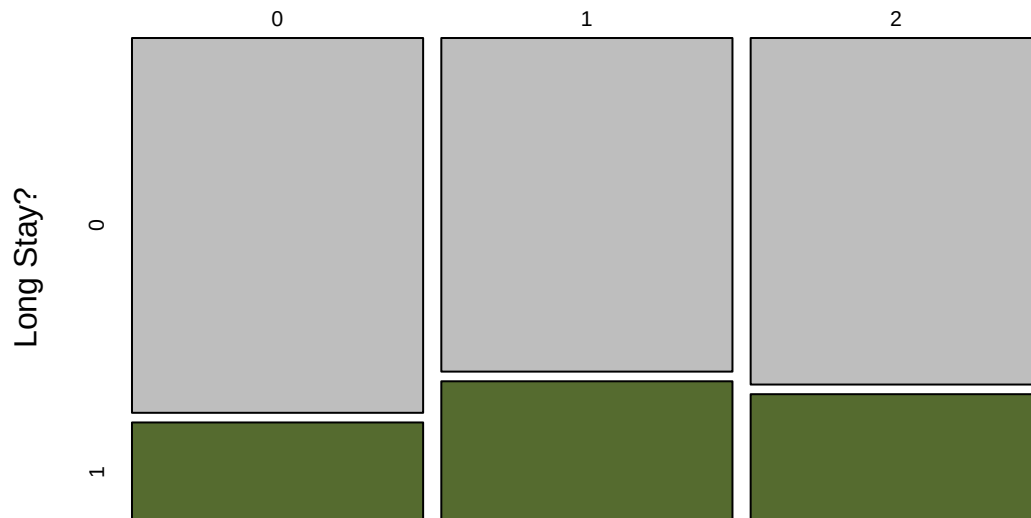
Decades vs Long-stay Rate



There are notably fewer long-stays in the 60s, 70s, and 80s than surrounding years, with
 ### another notably low percentage of long-stays in the 2020 decade. There isn't a linear trend
 ### so we shouldn't treat this variable as continuous.

```
plot(tprint("Artist.Structure"), main="Long-stay rate vs Artist Structure", ylab="Long Stay?",
      col=c("gray", "darkolivegreen"))
```

Long-stay rate vs Artist Structure



Long-stays are the least prevalent for groups with three or more people,
somewhat higher for duos, and slightly higher yet for individuals (about 10% higher than for
trios or larger).

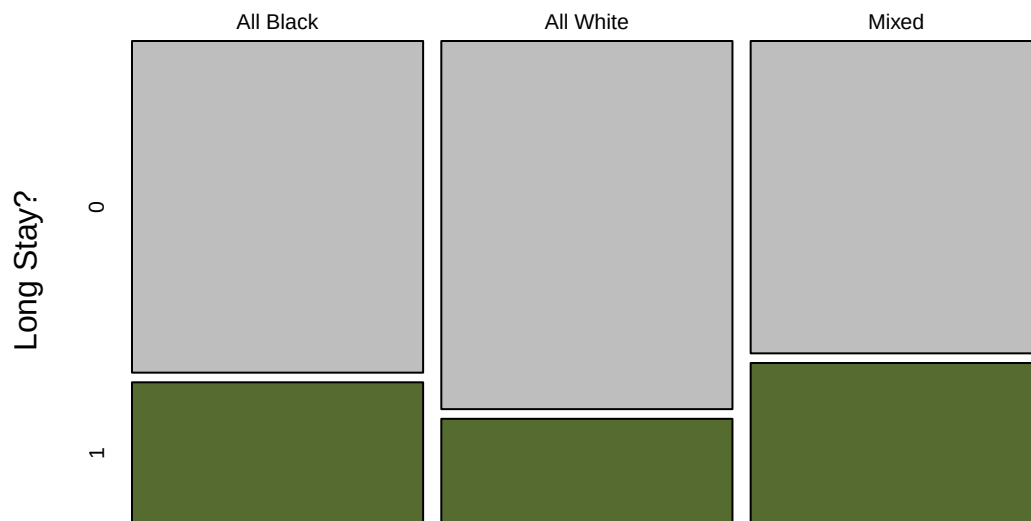
```
plot(tprint("Artist.Male"), main="Long-stay Rate vs Male", ylab="Long Stay?",  
     col=c("gray", "darkolivegreen"))
```



Mixed-gender groups have the highest percentage of long-stay songs compared to individuals
 ### and single-gender groups. Females have the lowest long-stay percentage and males are in
 ### the middle.

```
plot(tprint("Race"), main="Long-stay Rate vs Race", ylab="Long Stay?",
     col=c("gray", "darkolivegreen"))
```

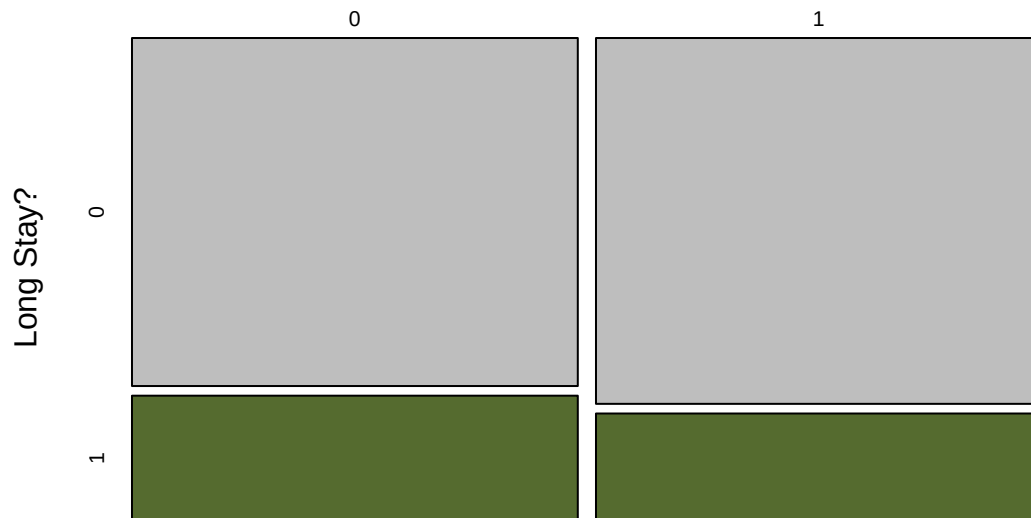
Long-stay Rate vs Race



Mixed-race groups have the highest percentage of long-stay songs, with black groups/individuals
in second and white groups/individuals with the lowest percentage.

```
plot(tprint("Vocally.Based"), main="Long-stay Rate vs Vocal-based", ylab="Long Stay?",  
     col=c("gray", "darkolivegreen"))
```

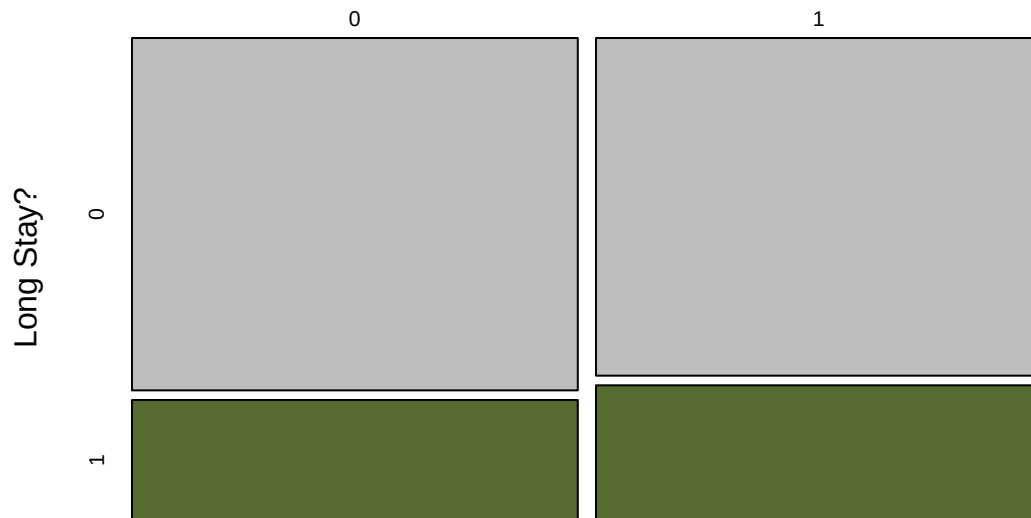
Long-stay Rate vs Vocal-based



The long-stay rate is similar for vocally-based and non-vocally-based songs, though it
is higher for non-vocally based songs.

```
plot(tprint("Bass.Based"), main="Long-stay Rate vs Bass-based", ylab="Long Stay?",  
     col=c("gray", "darkolivegreen"))
```

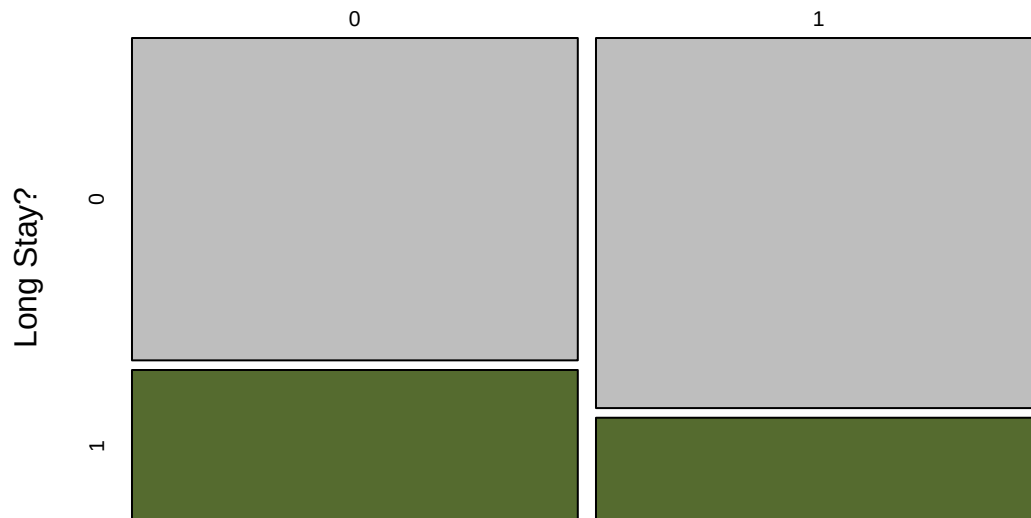
Long-stay Rate vs Bass-based



The long-stay rate is slightly higher for bass-based songs.

```
plot(tprint("Guitar.Based"), main="Long-stay Rate vs Guitar-based", ylab="Long Stay?",  
     col=c("gray", "darkolivegreen"))
```

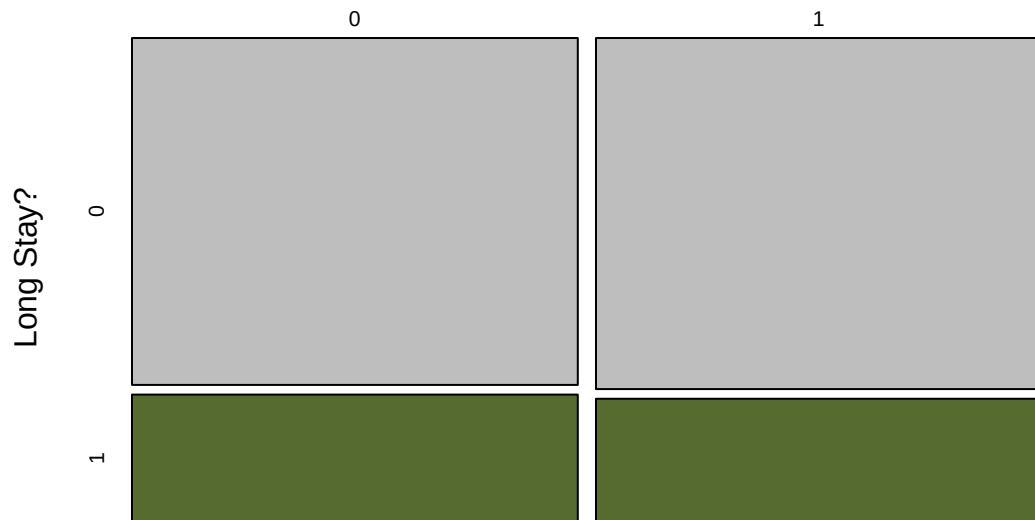
Long-stay Rate vs Guitar-based



The long-stay rate is considerably lower for guitar-based songs.

```
plot(tprint("Piano.Based"), main="Long-stay Rate vs Piano-based", ylab="Long Stay?",  
col=c("gray", "darkolivegreen"))
```

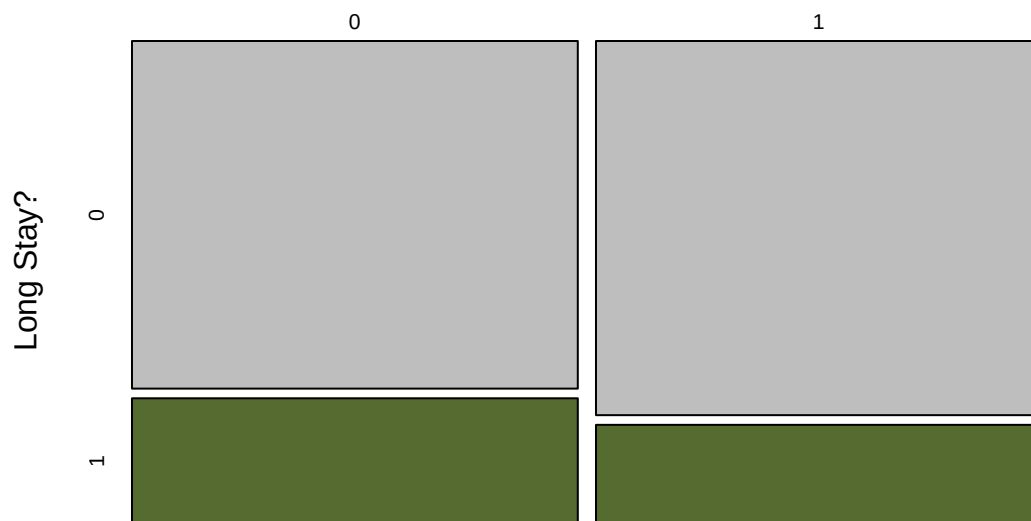
Long-stay Rate vs Piano-based



The long-stay rate is very similar but slightly higher for non-piano-based songs.

```
plot(tprint("Saxophone"), main="Long-stay Rate vs Saxophone", ylab="Long Stay?",  
     col=c("gray", "darkolivegreen"))
```

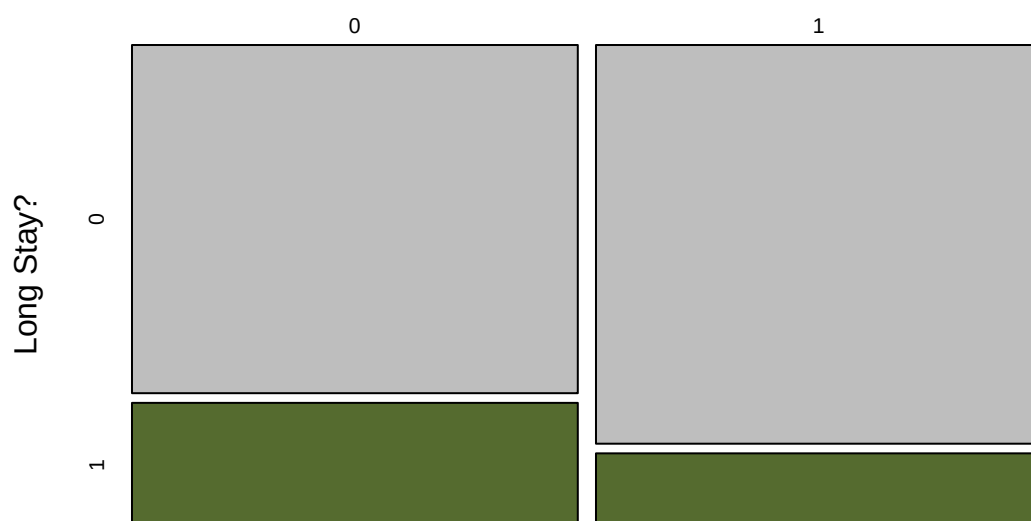
Long-stay Rate vs Saxophone



The long-stay rate is somewhat lower for songs that feature saxophone.

```
plot(tprint("Trumpet"), main="Long-stay Rate vs Trumpet", ylab="Long Stay?",  
     col=c("gray", "darkolivegreen"))
```

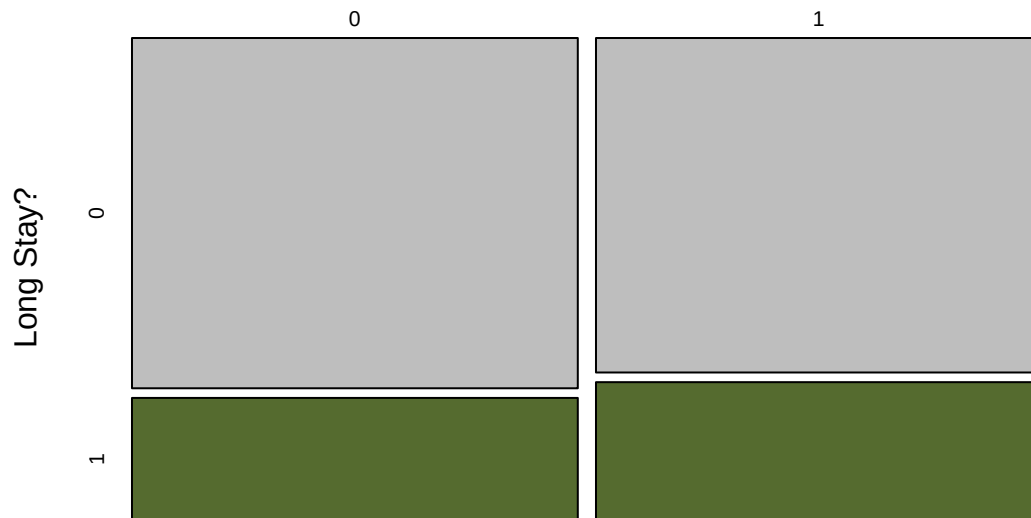
Long-stay Rate vs Trumpet



The long-stay rate is considerably lower for songs that feature trumpet.

```
plot(tprint("Written.for.a.Film"), main="Long-stay Rate vs Written for a Film",  
      ylab="Long Stay?", col=c("gray", "darkolivegreen"))
```

Long-stay Rate vs Written for a Film



The long-stay rate is slightly higher for songs that were written for films.

Continuous vars

Length

```
cats <- c(min(Length)-1, 150, 180, 210, 240, 270, 300, max(Length))
```

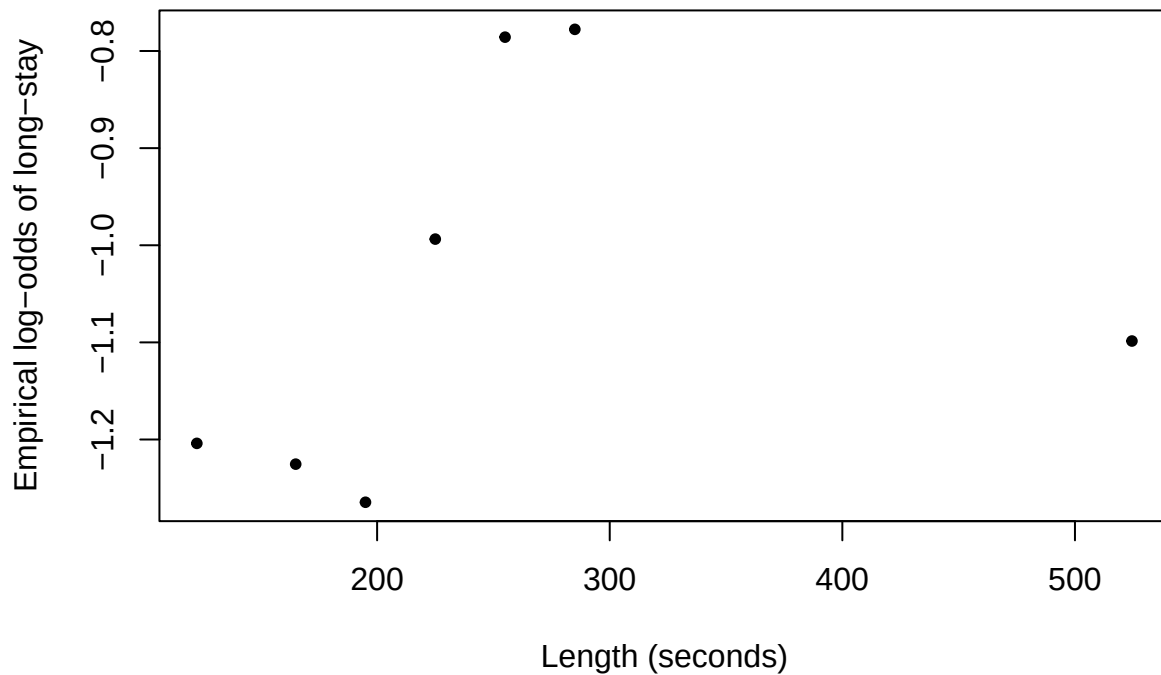
```
emp_probs <- numeric(length(cats)-1)
```

```
n <- numeric(length(cats)-1)
```

```
for (i in 1:(length(cats)-1)){
  emp_probs[i] <- mean(Long.Stay[Length > cats[i] & Length <= cats[i+1]])
  emp_probs[i] <- log(emp_probs[i]/(1-emp_probs[i]))
  n[i] <- sum(Length > cats[i] & Length <= cats[i+1])
}
```

```
to_plot <- (cats[-length(cats)] + cats[-1])/2 # average value for each category
```

```
plot(to_plot, emp_probs, pch=20, xlab = "Length (seconds)", ylab="Empirical log-odds of long-stay")
```

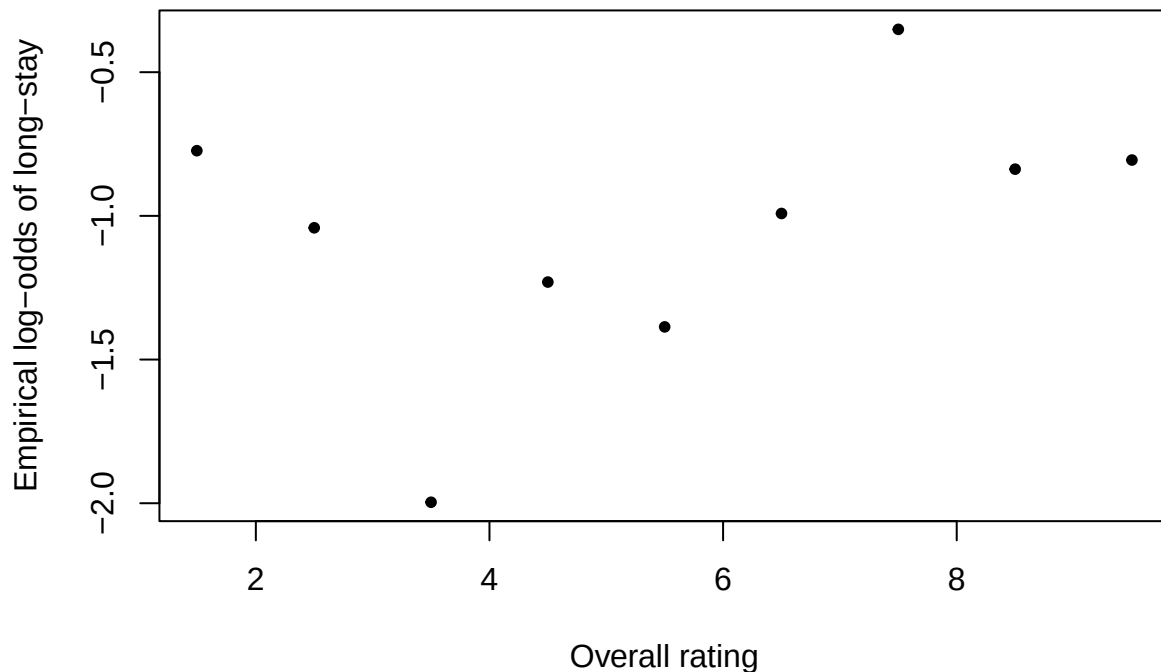


```
### There isn't a clear trend between length and log-odds of being a long-stay, though it is clear
### that the empirical probability is lower for songs shorter than 200 seconds, and there is a peak
### for songs just under 300 seconds. Longer songs (over 300 seconds) again have a low empirical
### probability of being a long-stay.
```

```
# Overall Rating
cats <- c(0.99, 2:10)
emp_probs <- numeric(length(cats)-1)
n <- numeric(length(cats)-1)

for (i in 1:(length(cats)-1)){
  emp_probs[i] <- mean(Long.Stay[Overall.Rating > cats[i] & Overall.Rating <= cats[i+1]])
  emp_probs[i] <- log(emp_probs[i]/(1-emp_probs[i]))
  n[i] <- sum(Overall.Rating > cats[i] & Overall.Rating <= cats[i+1])
}

to_plot <- (cats[-length(cats)] + cats[-1])/2 # average value for each category
plot(to_plot, emp_probs, pch=20, xlab = "Overall rating", ylab="Empirical log-odds of long-stay")
```



```
### There isn't a clear trend between overall rating and log-odds of being a long-stay, though the
### empirical probability is particularly low for ratings between 3 and 4 and particularly high for
### ratings between 7 and 8. The other data points seem relatively horizontal, showing no obvious
### trend.
```

```
# Disagreement
```

```
# There is an extreme value here (16.6), and this song was not a long-stay. The plot focuses on the
# other observations.
```

```
cats <- c(-0.1, 0.67, 1.33, 2, 2.67, 3.33, 6)-0.01
```

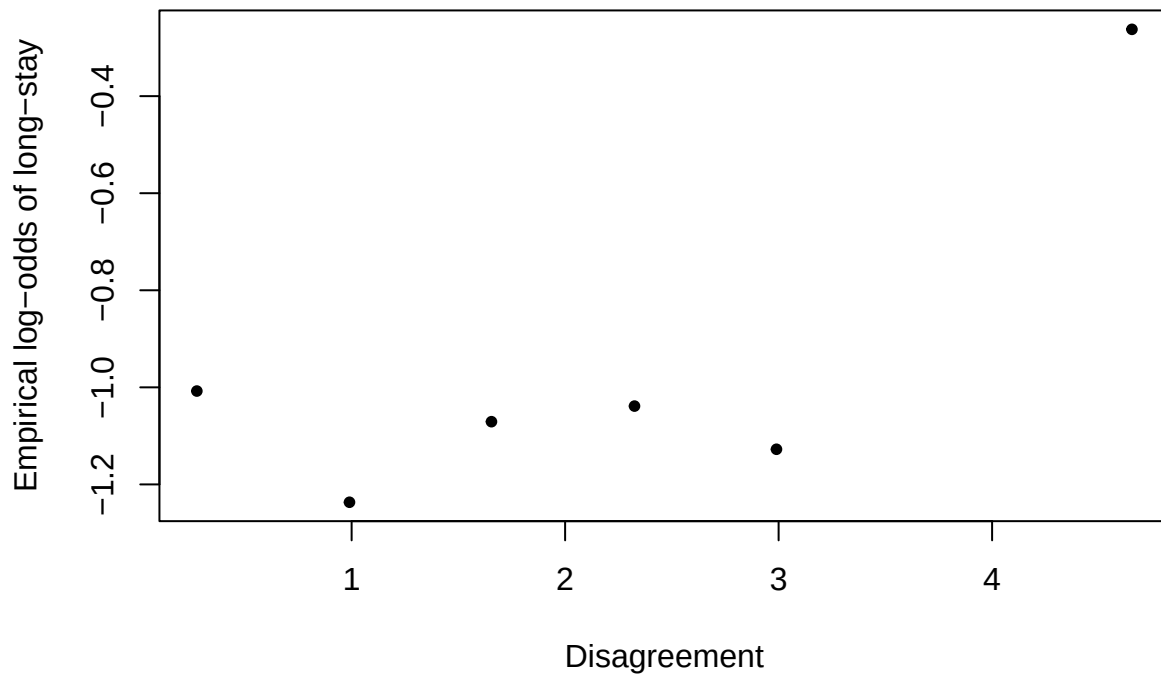
```
emp_probs <- numeric(length(cats)-1)
```

```
n <- numeric(length(cats)-1)
```

```
for (i in 1:(length(cats)-1)){
  emp_probs[i] <- mean(Long.Stay[Disagreement > cats[i] & Disagreement <= cats[i+1]])
  emp_probs[i] <- log(emp_probs[i]/(1-emp_probs[i]))
  n[i] <- sum(Disagreement > cats[i] & Disagreement <= cats[i+1])
}
```

```
to_plot <- (cats[-length(cats)] + cats[-1])/2 # average value for each category
```

```
plot(to_plot, emp_probs, pch=20, xlab = "Disagreement", ylab="Empirical log-odds of long-stay")
```

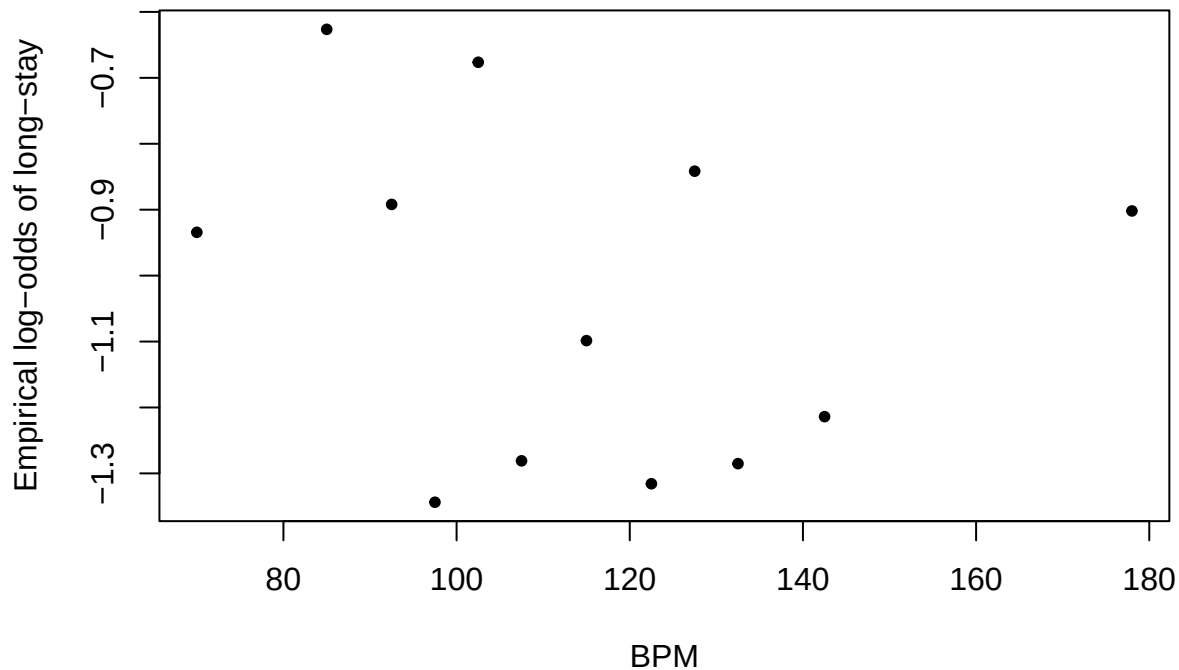


*# There is no clear pattern, but there is a much higher empirical log-odds of being a long-stay for
Disagreement scores over 3.33 compared to disagreement scores below 3.33.*

```
# BPM
cats <- c(60, 80, 90, 95, 100, 105, 110, 120, 125, 130, 135, 150, max(BPM))
emp_probs <- numeric(length(cats)-1)
n <- numeric(length(cats)-1)

for (i in 1:(length(cats)-1)){
  emp_probs[i] <- mean(Long.Stay[BPM > cats[i] & BPM <= cats[i+1]])
  emp_probs[i] <- log(emp_probs[i]/(1-emp_probs[i]))
  n[i] <- sum(BPM > cats[i] & BPM <= cats[i+1])
}

to_plot <- (cats[-length(cats)] + cats[-1])/2 # average value for each category
plot(to_plot, emp_probs, pch=20, xlab = "BPM", ylab="Empirical log-odds of long-stay")
```

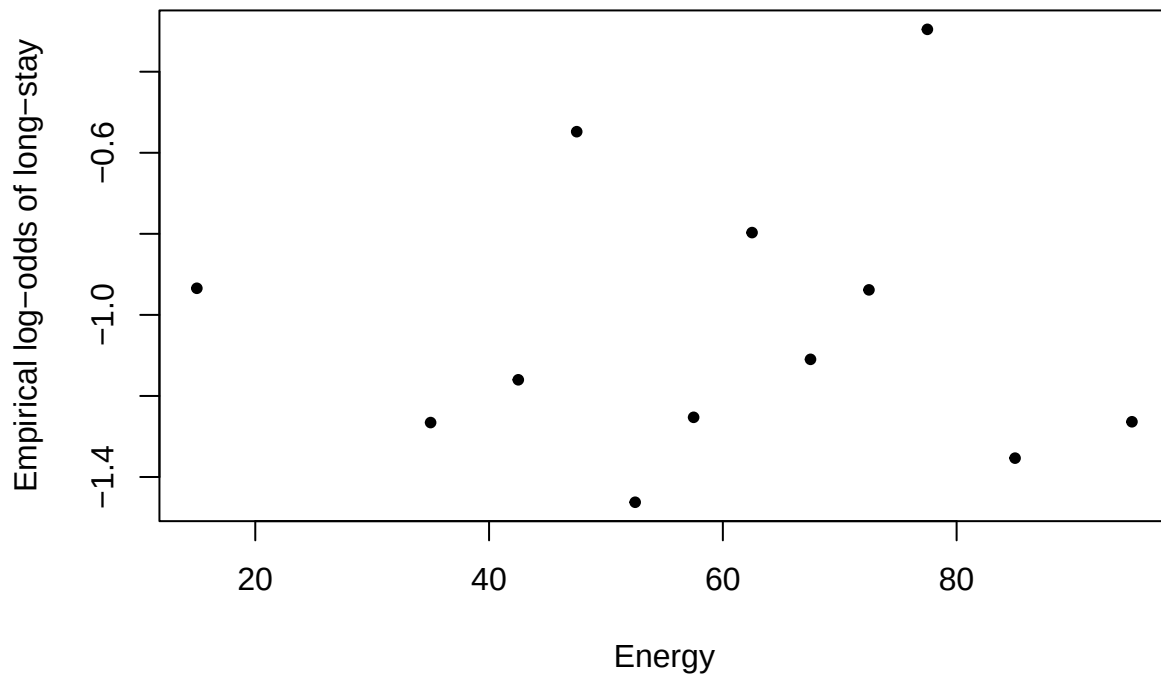


There seems to be a negative association between BPM and log-odds of being a long-stay from 60 to # 150, but at higher BPMs (>150) there is an increased log-odds of being a long-stay. This is likely # not a result of small sample size in the last category (n=97), so the effect may not be linear.

```
# Energy
cats <- c(0, 30, 40, 45, 50, 55, 60, 65, 70, 75, 80, 90, 100)
emp_probs <- numeric(length(cats)-1)
n <- numeric(length(cats)-1)

for (i in 1:(length(cats)-1)){
  emp_probs[i] <- mean(Long.Stay[Energy > cats[i] & Energy <= cats[i+1]])
  emp_probs[i] <- log(emp_probs[i]/(1-emp_probs[i]))
  n[i] <- sum(Energy > cats[i] & Energy <= cats[i+1])
}

to_plot <- (cats[-length(cats)] + cats[-1])/2 # average value for each category
plot(to_plot, emp_probs, pch=20, xlab = "Energy", ylab="Empirical log-odds of long-stay")
```

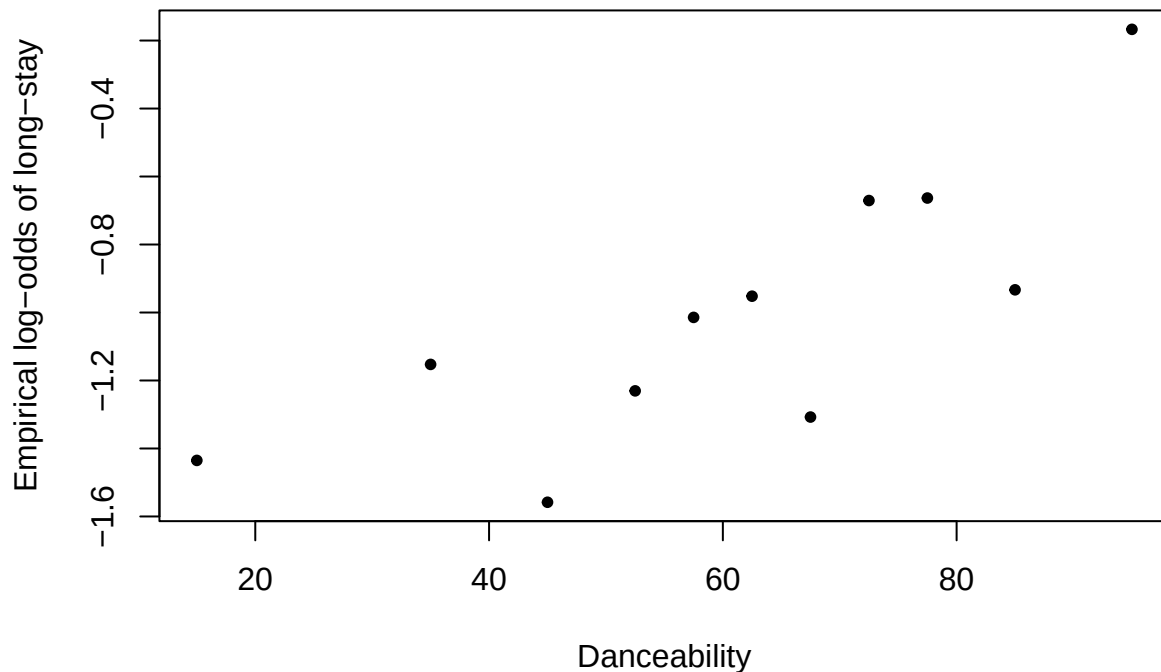


The empirical log-odds are all over the place for different categories of energy and there is no obvious trend.

```
# Danceability
cats <- c(0, 30, 40, 50, 55, 60, 65, 70, 75, 80, 90, 100)
emp_probs <- numeric(length(cats)-1)
n <- numeric(length(cats)-1)

for (i in 1:(length(cats)-1)){
  emp_probs[i] <- mean(Long.Stay[Danceability > cats[i] & Danceability <= cats[i+1]])
  emp_probs[i] <- log(emp_probs[i]/(1-emp_probs[i]))
  n[i] <- sum(Danceability > cats[i] & Danceability <= cats[i+1])
}

to_plot <- (cats[-length(cats)] + cats[-1])/2 # average value for each category
plot(to_plot, emp_probs, pch=20, xlab = "Danceability", ylab="Empirical log-odds of long-stay")
```

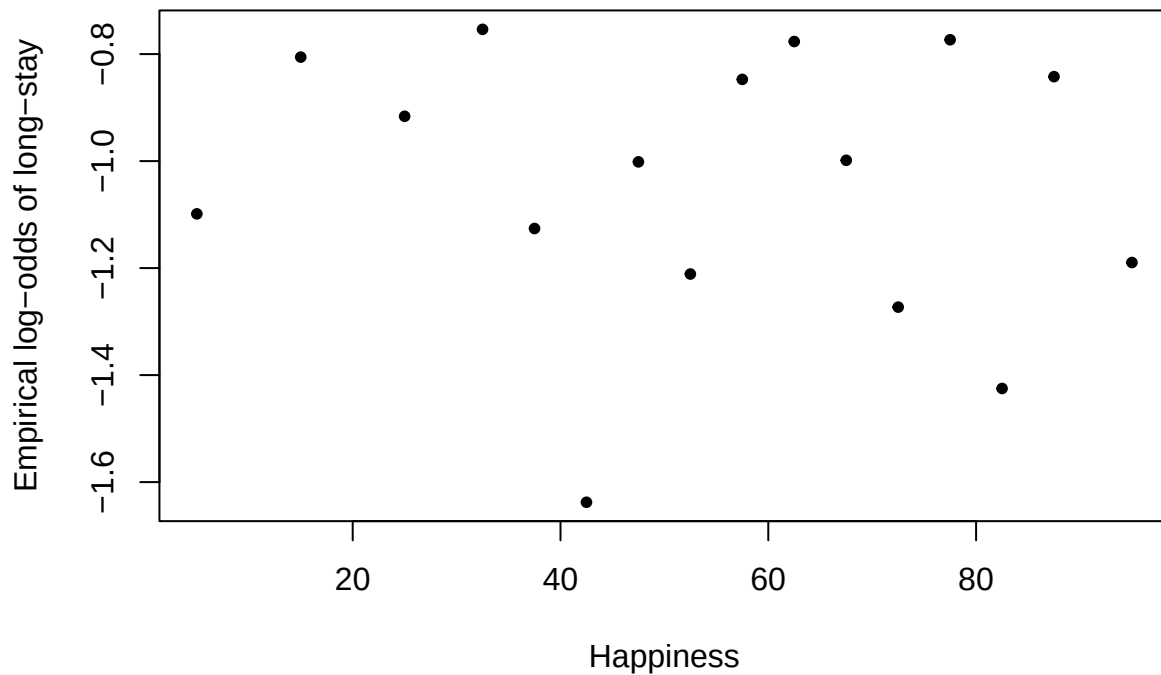


*# The relationship between empirical log-odds and danceability isn't particularly strong for songs
with low danceabilities, but there seems to be a positive relationship between danceability and
empirical log-odds of being a long-stay for songs with danceabilities over 40.*

```
# Happiness
cats <- sort(c(0:10*10, 3.5:8.5*10), F)
emp_probs <- numeric(length(cats)-1)
n <- numeric(length(cats)-1)

for (i in 1:(length(cats)-1)){
  emp_probs[i] <- mean(Long.Stay[Happiness > cats[i] & Happiness <= cats[i+1]])
  emp_probs[i] <- log(emp_probs[i]/(1-emp_probs[i]))
  n[i] <- sum(Happiness > cats[i] & Happiness <= cats[i+1])
}

to_plot <- (cats[-length(cats)] + cats[-1])/2 # average value for each category
plot(to_plot, emp_probs, pch=20, xlab = "Happiness", ylab="Empirical log-odds of long-stay")
```

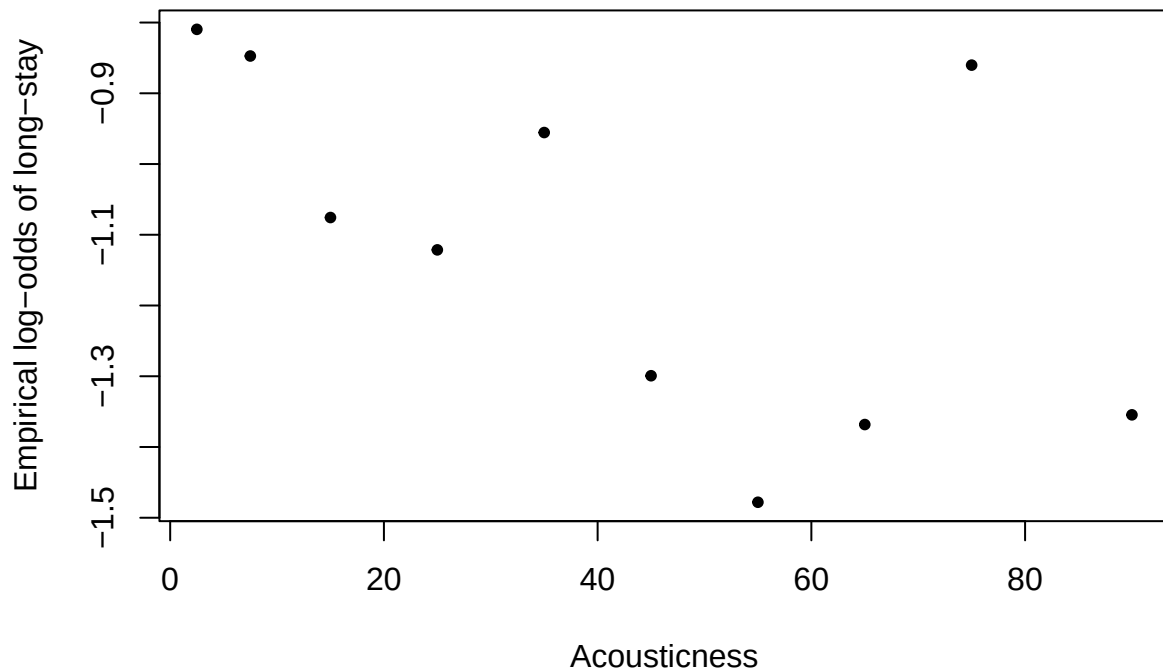


The relationship between empirical log-odds and danceability isn't particularly strong.

```
# Acousticness
cats <- c(-0.01, 5, 10, 20, 30, 40, 50, 60, 70, 80, 100)
emp_probs <- numeric(length(cats)-1)
n <- numeric(length(cats)-1)

for (i in 1:(length(cats)-1)){
  emp_probs[i] <- mean(Long.Stay[Acousticness > cats[i] & Acousticness <= cats[i+1]])
  emp_probs[i] <- log(emp_probs[i]/(1-emp_probs[i]))
  n[i] <- sum(Acousticness > cats[i] & Acousticness <= cats[i+1])
}

to_plot <- (cats[-length(cats)] + cats[-1])/2 # average value for each category
plot(to_plot, emp_probs, pch=20, xlab = "Acousticness", ylab="Empirical log-odds of long-stay")
```



*# The relationship isn't particularly strong, but there could be a negative relationship between the
empirical log-odds of being a long-stay and and the probability of being an acoustic song
(according to Spotify).*

```
interested_vars <- c("Decade", "Overall.Rating", "Disagreement", "Artist.Structure", "BPM", "Energy", "I")
pvals <- numeric(length(interested_vars))
```

```
for (i in seq_along(interested_vars)) {

  varname <- interested_vars[i]
  cat("Variable:", varname, "\n")

  #formula
  fmla <- as.formula(paste("Long.Stay ~", varname))

  m <- glm(fmla, data = data, family = "binomial")
  print(summary(m))

  #extract p-values
  pvals[i] <- summary(m)$coefficients[2, 4]
}
```

```
## Variable: Decade
##
```

```

## Call:
## glm(formula = fmla, family = "binomial", data = data)
##
## Coefficients:
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept) -40.472167   6.982708  -5.796 6.79e-09 ***
## Decade       0.019873   0.003516   5.653 1.58e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##    Null deviance: 1350.1  on 1171  degrees of freedom
## Residual deviance: 1317.7  on 1170  degrees of freedom
## AIC: 1321.7
##
## Number of Fisher Scoring iterations: 4
##
## Variable: Overall.Rating
##
## Call:
## glm(formula = fmla, family = "binomial", data = data)
##
## Coefficients:
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept)  -1.90189   0.24617  -7.726 1.11e-14 ***
## Overall.Rating 0.13940   0.03731   3.736 0.000187 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##    Null deviance: 1350.1  on 1171  degrees of freedom
## Residual deviance: 1335.8  on 1170  degrees of freedom
## AIC: 1339.8
##
## Number of Fisher Scoring iterations: 4
##
## Variable: Disagreement
##
## Call:
## glm(formula = fmla, family = "binomial", data = data)
##
## Coefficients:
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept)  -1.22023   0.11954 -10.208  <2e-16 ***
## Disagreement  0.11838   0.06137   1.929   0.0537 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##    Null deviance: 1350.1  on 1171  degrees of freedom
## Residual deviance: 1346.3  on 1170  degrees of freedom

```

```

## AIC: 1350.3
##
## Number of Fisher Scoring iterations: 4
##
## Variable: Artist.Structure
##
## Call:
## glm(formula = fmla, family = "binomial", data = data)
##
## Coefficients:
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept)   -1.2416    0.1091 -11.383  <2e-16 ***
## Artist.Structure  0.2771    0.1106   2.506   0.0122 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##    Null deviance: 1350.1  on 1171  degrees of freedom
## Residual deviance: 1343.8  on 1170  degrees of freedom
## AIC: 1347.8
##
## Number of Fisher Scoring iterations: 4
##
## Variable: BPM
##
## Call:
## glm(formula = fmla, family = "binomial", data = data)
##
## Coefficients:
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept) -0.869679    0.321614  -2.704   0.00685 **
## BPM         -0.001398    0.002723  -0.513   0.60778
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##    Null deviance: 1350.1  on 1171  degrees of freedom
## Residual deviance: 1349.8  on 1170  degrees of freedom
## AIC: 1353.8
##
## Number of Fisher Scoring iterations: 4
##
## Variable: Energy
##
## Call:
## glm(formula = fmla, family = "binomial", data = data)
##
## Coefficients:
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept) -1.0099181   0.2183930  -4.624 3.76e-06 ***
## Energy      -0.0003565   0.0034434  -0.104   0.918
## ---

```

```

## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 1350.1  on 1171  degrees of freedom
## Residual deviance: 1350.0  on 1170  degrees of freedom
## AIC: 1354
##
## Number of Fisher Scoring iterations: 4
##
## Variable: Danceability
##
## Call:
## glm(formula = fmla, family = "binomial", data = data)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)  -1.929682   0.297800  -6.480 9.18e-11 ***
## Danceability   0.014092   0.004499   3.132 0.00174 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 1350.1  on 1171  degrees of freedom
## Residual deviance: 1340.0  on 1170  degrees of freedom
## AIC: 1344
##
## Number of Fisher Scoring iterations: 4
##
## Variable: Happiness
##
## Call:
## glm(formula = fmla, family = "binomial", data = data)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)  -0.866352   0.175593  -4.934 8.06e-07 ***
## Happiness    -0.002690   0.002665  -1.009   0.313
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 1350.1  on 1171  degrees of freedom
## Residual deviance: 1349.0  on 1170  degrees of freedom
## AIC: 1353
##
## Number of Fisher Scoring iterations: 4
##
## Variable: Acousticness
##
## Call:
## glm(formula = fmla, family = "binomial", data = data)

```

```

##
## Coefficients:
##           Estimate Std. Error z value Pr(>|z|)
## (Intercept) -0.863482   0.096341  -8.963   <2e-16 ***
## Acousticness -0.005812   0.002503  -2.322   0.0202 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##    Null deviance: 1350.1  on 1171  degrees of freedom
## Residual deviance: 1344.5  on 1170  degrees of freedom
## AIC: 1348.5
##
## Number of Fisher Scoring iterations: 4
##
## Variable: Vocally.Based
##
## Call:
## glm(formula = fmla, family = "binomial", data = data)
##
## Coefficients:
##           Estimate Std. Error z value Pr(>|z|)
## (Intercept)  -1.02056    0.06814 -14.978   <2e-16 ***
## Vocally.Based -0.20322    0.30153  -0.674     0.5
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##    Null deviance: 1350.1  on 1171  degrees of freedom
## Residual deviance: 1349.6  on 1170  degrees of freedom
## AIC: 1353.6
##
## Number of Fisher Scoring iterations: 4
##
## Variable: Bass.Based
##
## Call:
## glm(formula = fmla, family = "binomial", data = data)
##
## Coefficients:
##           Estimate Std. Error z value Pr(>|z|)
## (Intercept)  -1.0680     0.0761 -14.033   <2e-16 ***
## Bass.Based    0.1570     0.1557   1.008   0.313
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##    Null deviance: 1350.1  on 1171  degrees of freedom
## Residual deviance: 1349.1  on 1170  degrees of freedom
## AIC: 1353.1
##

```

```

## Number of Fisher Scoring iterations: 4
##
## Variable: Guitar.Based
##
## Call:
## glm(formula = fmla, family = "binomial", data = data)
##
## Coefficients:
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept)  -0.75769    0.09435  -8.031 9.69e-16 ***
## Guitar.Based -0.51779    0.13360  -3.876 0.000106 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##    Null deviance: 1350.1  on 1171  degrees of freedom
## Residual deviance: 1335.0  on 1170  degrees of freedom
## AIC: 1339
##
## Number of Fisher Scoring iterations: 4
##
## Variable: Piano.Based
##
## Call:
## glm(formula = fmla, family = "binomial", data = data)
##
## Coefficients:
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept)  -1.00866    0.09383 -10.750  <2e-16 ***
## Piano.Based  -0.04540    0.13273  -0.342   0.732
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##    Null deviance: 1350.1  on 1171  degrees of freedom
## Residual deviance: 1349.9  on 1170  degrees of freedom
## AIC: 1353.9
##
## Number of Fisher Scoring iterations: 4
##
## Variable: Saxophone
##
## Call:
## glm(formula = fmla, family = "binomial", data = data)
##
## Coefficients:
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept)  -1.01621    0.06795 -14.956  <2e-16 ***
## Saxophone    -0.31066    0.31929  -0.973   0.331
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##

```

```

## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 1350.1  on 1171  degrees of freedom
## Residual deviance: 1349.1  on 1170  degrees of freedom
## AIC: 1353.1
##
## Number of Fisher Scoring iterations: 4
##
## Variable: Trumpet
##
## Call:
## glm(formula = fmla, family = "binomial", data = data)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept) -1.02257    0.06677  -15.31  <2e-16 ***
## Trumpet      -0.65141    0.63269   -1.03    0.303
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 1350.1  on 1171  degrees of freedom
## Residual deviance: 1348.8  on 1170  degrees of freedom
## AIC: 1352.8
##
## Number of Fisher Scoring iterations: 4
##
## Variable: Written.for.a.Film
##
## Call:
## glm(formula = fmla, family = "binomial", data = data)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)   -1.04501    0.06936  -15.067  <2e-16 ***
## Written.for.a.Film 0.16646    0.23923    0.696    0.487
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 1350.1  on 1171  degrees of freedom
## Residual deviance: 1349.6  on 1170  degrees of freedom
## AIC: 1353.6
##
## Number of Fisher Scoring iterations: 4
##
## Variable: Length
##
## Call:
## glm(formula = fmla, family = "binomial", data = data)
##
## Coefficients:

```

```

##           Estimate Std. Error z value Pr(>|z|)
## (Intercept) -1.596893   0.283999  -5.623 1.88e-08 ***
## Length      0.002532   0.001228   2.061  0.0393 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##    Null deviance: 1350.1  on 1171  degrees of freedom
## Residual deviance: 1345.8  on 1170  degrees of freedom
## AIC: 1349.8
##
## Number of Fisher Scoring iterations: 4
##
## Variable: Race
##
## Call:
## glm(formula = fmla, family = "binomial", data = data)
##
## Coefficients:
##           Estimate Std. Error z value Pr(>|z|)
## (Intercept)  -0.8497     0.1100  -7.725 1.11e-14 ***
## RaceAll White -0.4020     0.1457  -2.760  0.00578 **
## RaceMixed     0.1877     0.2074   0.905  0.36566
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##    Null deviance: 1350.1  on 1171  degrees of freedom
## Residual deviance: 1337.6  on 1169  degrees of freedom
## AIC: 1343.6
##
## Number of Fisher Scoring iterations: 4

```

pvals

```

## [1] 1.576417e-08 1.868238e-04 5.374337e-02 1.221882e-02 6.077761e-01
## [6] 9.175321e-01 1.736709e-03 3.128133e-01 2.022374e-02 5.003344e-01
## [11] 3.133871e-01 1.062531e-04 7.323377e-01 3.305748e-01 3.032020e-01
## [16] 4.865584e-01 3.926950e-02 5.780475e-03

```

```

results_table <- data.frame(
  Variable = interested_vars,
  P_Value = pvals
)

```

results_table

```

##           Variable      P_Value
## 1           Decade 1.576417e-08
## 2 Overall.Rating 1.868238e-04
## 3    Disagreement 5.374337e-02
## 4 Artist.Structure 1.221882e-02
## 5              BPM 6.077761e-01

```

```
## 6          Energy 9.175321e-01
## 7      Danceability 1.736709e-03
## 8          Happiness 3.128133e-01
## 9      Acousticness 2.022374e-02
## 10     Vocally.Based 5.003344e-01
## 11          Bass.Based 3.133871e-01
## 12     Guitar.Based 1.062531e-04
## 13     Piano.Based 7.323377e-01
## 14          Saxophone 3.305748e-01
## 15          Trumpet 3.032020e-01
## 16 Written.for.a.Film 4.865584e-01
## 17          Length 3.926950e-02
## 18          Race 5.780475e-03
```

```
## When fitting individual variables, the ones that are significant at an alpha = 0.05 level are:
# Decade, Overall.Rating, Artist.StructureSolo, Danceability, Acousticness, Guitar.Based, Length,
# RaceAllWhite.
```

Model Analysis

Start out by initially fitting the full model with all “reasonable” predictors. We will denote this by m_0 . We will iteratively test the coefficient with the highest p-value for significance in the model using a Likelihood Ratio Test (Deviance) and check the AIC to compare the $m(n)$ and $m(n+1)$ model.

```
library(kableExtra)

coef_df <- data.frame(
  Predictor = c(
    "Decade", "Overall.Rating", "Artist.Structure", "BPM",
    "Energy", "Danceability", "Happiness", "Acousticness",
    "Vocally.Based", "Bass.Based", "Guitar.Based", "Piano.Based",
    "Saxophone", "Trumpet", "Written.for.a.Film", "Length", "Race"
  ),
  `p-value` = c(
    "<0.001***", "<0.001***", "0.203", "0.608",
    "0.918", "0.002**", "0.313", "0.020*",
    "0.500", "0.313", "<0.001***", "0.732",
    "0.331", "0.303", "0.487", "0.039*", "0.006**"
  )
)

kable(coef_df,
      format = "latex",
      booktabs = TRUE,
      caption = "P-values for Predictors in the Individual Logistic Regression Model",
      align = "l c") %>%
  kable_styling(latex_options = "hold_position")

billboard <- data
m0 <- glm(
  Long.Stay ~ BPM + Energy + Danceability + Happiness + Acousticness +
    Length + Overall.Rating + Disagreement +
    Artist.Male + Artist.White + Artist.Black +
    Vocally.Based + Bass.Based + Guitar.Based + Piano.Based +
    Saxophone + Trumpet +
```

Table 3: P-values for Predictors in the Individual Logistic Regression Model

Predictor	p.value
Decade	<0.001***
Overall.Rating	<0.001***
Artist.Structure	0.203
BPM	0.608
Energy	0.918
Danceability	0.002**
Happiness	0.313
Acousticness	0.020*
Vocally.Based	0.500
Bass.Based	0.313
Guitar.Based	<0.001***
Piano.Based	0.732
Saxophone	0.331
Trumpet	0.303
Written.for.a.Film	0.487
Length	0.039*
Race	0.006**

```

    Written.for.a.Film + Artist.Structure + Decade,
    family = binomial,
    data = billboard
)
summary(m0)

```

```

##
## Call:
## glm(formula = Long.Stay ~ BPM + Energy + Danceability + Happiness +
##      Acousticness + Length + Overall.Rating + Disagreement + Artist.Male +
##      Artist.White + Artist.Black + Vocally.Based + Bass.Based +
##      Guitar.Based + Piano.Based + Saxophone + Trumpet + Written.for.a.Film +
##      Artist.Structure + Decade, family = binomial, data = billboard)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)   -3.424e+01  9.896e+00  -3.460 0.000541 ***
## BPM           -6.405e-04  2.868e-03  -0.223 0.823299
## Energy        -9.185e-03  4.940e-03  -1.859 0.063001 .
## Danceability    6.276e-03  6.005e-03   1.045 0.295931
## Happiness      1.241e-03  3.911e-03   0.317 0.751023
## Acousticness   -2.312e-03  3.438e-03  -0.673 0.501223
## Length         4.242e-04  1.409e-03   0.301 0.763436
## Overall.Rating  2.087e-01  4.217e-02   4.949 7.46e-07 ***
## Disagreement   1.222e-01  6.422e-02   1.903 0.057062 .
## Artist.Male    1.389e-01  1.111e-01   1.250 0.211354
## Artist.White   1.550e-01  1.131e-01   1.370 0.170532
## Artist.Black   1.072e-01  1.115e-01   0.961 0.336561
## Vocally.Based  1.558e-01  3.258e-01   0.478 0.632638

```

```

## Bass.Based      -5.959e-03  1.766e-01  -0.034  0.973089
## Guitar.Based    -3.451e-01  1.643e-01  -2.101  0.035636 *
## Piano.Based     -2.773e-03  1.534e-01  -0.018  0.985578
## Saxophone       -1.477e-01  3.340e-01  -0.442  0.658299
## Trumpet         -6.829e-01  6.479e-01  -1.054  0.291843
## Written.for.a.Film 1.878e-01  2.493e-01   0.753  0.451215
## Artist.Structure 1.254e-01  1.224e-01   1.024  0.305793
## Decade          1.593e-02  4.991e-03   3.192  0.001413 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
## Null deviance: 1350.1 on 1171 degrees of freedom
## Residual deviance: 1274.5 on 1151 degrees of freedom
## AIC: 1316.5
##
## Number of Fisher Scoring iterations: 4
m1 <- update(m0, . ~ . - Artist.Structure)
anova(m0, m1, test = "LRT")

## Analysis of Deviance Table
##
## Model 1: Long.Stay ~ BPM + Energy + Danceability + Happiness + Acousticness +
## Length + Overall.Rating + Disagreement + Artist.Male + Artist.White +
## Artist.Black + Vocally.Based + Bass.Based + Guitar.Based +
## Piano.Based + Saxophone + Trumpet + Written.for.a.Film +
## Artist.Structure + Decade
## Model 2: Long.Stay ~ BPM + Energy + Danceability + Happiness + Acousticness +
## Length + Overall.Rating + Disagreement + Artist.Male + Artist.White +
## Artist.Black + Vocally.Based + Bass.Based + Guitar.Based +
## Piano.Based + Saxophone + Trumpet + Written.for.a.Film +
## Decade
## Resid. Df Resid. Dev Df Deviance Pr(>Chi)
## 1      1151      1274.5
## 2      1152      1275.6 -1   -1.0456   0.3065

summary(m1)

##
## Call:
## glm(formula = Long.Stay ~ BPM + Energy + Danceability + Happiness +
## Acousticness + Length + Overall.Rating + Disagreement + Artist.Male +
## Artist.White + Artist.Black + Vocally.Based + Bass.Based +
## Guitar.Based + Piano.Based + Saxophone + Trumpet + Written.for.a.Film +
## Decade, family = binomial, data = billboard)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)  -3.652e+01  9.645e+00  -3.786 0.000153 ***
## BPM          -7.891e-04  2.863e-03  -0.276 0.782871
## Energy       -9.240e-03  4.939e-03  -1.871 0.061397 .
## Danceability  6.263e-03  6.009e-03   1.042 0.297331
## Happiness    1.140e-03  3.909e-03   0.292 0.770494

```

```

## Acousticness      -1.882e-03  3.412e-03  -0.552  0.581281
## Length            4.143e-04  1.405e-03   0.295  0.768107
## Overall.Rating    2.073e-01  4.214e-02   4.920  8.65e-07 ***
## Disagreement      1.250e-01  6.412e-02   1.949  0.051265 .
## Artist.Male       1.385e-01  1.112e-01   1.245  0.212973
## Artist.White      1.505e-01  1.128e-01   1.334  0.182164
## Artist.Black      1.037e-01  1.112e-01   0.933  0.350985
## Vocally.Based     1.457e-01  3.257e-01   0.447  0.654563
## Bass.Based        5.880e-03  1.763e-01   0.033  0.973391
## Guitar.Based     -3.561e-01  1.639e-01  -2.172  0.029837 *
## Piano.Based       3.211e-04  1.534e-01   0.002  0.998329
## Saxophone        -1.330e-01  3.335e-01  -0.399  0.689906
## Trumpet          -7.005e-01  6.489e-01  -1.079  0.280389
## Written.for.a.Film 1.961e-01  2.491e-01   0.787  0.431021
## Decade            1.714e-02  4.850e-03   3.534  0.000409 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
## Null deviance: 1350.1 on 1171 degrees of freedom
## Residual deviance: 1275.6 on 1152 degrees of freedom
## AIC: 1315.6
##
## Number of Fisher Scoring iterations: 4

```

```

m2 <- update(m1, . ~ . - Piano.Based)
anova(m1, m2, test = "LRT")

```

```

## Analysis of Deviance Table
##
## Model 1: Long.Stay ~ BPM + Energy + Danceability + Happiness + Acousticness +
## Length + Overall.Rating + Disagreement + Artist.Male + Artist.White +
## Artist.Black + Vocally.Based + Bass.Based + Guitar.Based +
## Piano.Based + Saxophone + Trumpet + Written.for.a.Film +
## Decade
## Model 2: Long.Stay ~ BPM + Energy + Danceability + Happiness + Acousticness +
## Length + Overall.Rating + Disagreement + Artist.Male + Artist.White +
## Artist.Black + Vocally.Based + Bass.Based + Guitar.Based +
## Saxophone + Trumpet + Written.for.a.Film + Decade
## Resid. Df Resid. Dev Df Deviance Pr(>Chi)
## 1      1152      1275.6
## 2      1153      1275.6 -1 -4.384e-06  0.9983

```

```
summary(m2)
```

```

##
## Call:
## glm(formula = Long.Stay ~ BPM + Energy + Danceability + Happiness +
## Acousticness + Length + Overall.Rating + Disagreement + Artist.Male +
## Artist.White + Artist.Black + Vocally.Based + Bass.Based +
## Guitar.Based + Saxophone + Trumpet + Written.for.a.Film +
## Decade, family = binomial, data = billboard)
##
## Coefficients:

```

```

##               Estimate Std. Error z value Pr(>|z|)
## (Intercept)    -3.651e+01  9.359e+00 -3.901 9.58e-05 ***
## BPM            -7.889e-04  2.862e-03 -0.276 0.782821
## Energy         -9.240e-03  4.939e-03 -1.871 0.061381 .
## Danceability    6.262e-03  6.009e-03  1.042 0.297300
## Happiness       1.140e-03  3.905e-03  0.292 0.770355
## Acousticness    -1.881e-03  3.411e-03 -0.552 0.581242
## Length          4.144e-04  1.405e-03  0.295 0.767984
## Overall.Rating  2.073e-01  4.211e-02  4.923 8.53e-07 ***
## Disagreement    1.250e-01  6.411e-02  1.949 0.051244 .
## Artist.Male     1.385e-01  1.112e-01  1.245 0.212954
## Artist.White    1.505e-01  1.128e-01  1.334 0.182110
## Artist.Black    1.037e-01  1.112e-01  0.933 0.350973
## Vocally.Based   1.457e-01  3.242e-01  0.449 0.653249
## Bass.Based      5.799e-03  1.720e-01  0.034 0.973105
## Guitar.Based    -3.562e-01  1.538e-01 -2.316 0.020533 *
## Saxophone       -1.330e-01  3.335e-01 -0.399 0.689910
## Trumpet         -7.005e-01  6.489e-01 -1.079 0.280378
## Written.for.a.Film 1.962e-01  2.486e-01  0.789 0.430108
## Decade          1.714e-02  4.721e-03  3.630 0.000283 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##    Null deviance: 1350.1  on 1171  degrees of freedom
## Residual deviance: 1275.6  on 1153  degrees of freedom
## AIC: 1313.6
##
## Number of Fisher Scoring iterations: 4
m3 <- update(m2, . ~ . - Bass.Based)
anova(m2, m3, test = "LRT")

## Analysis of Deviance Table
##
## Model 1: Long.Stay ~ BPM + Energy + Danceability + Happiness + Acousticness +
##   Length + Overall.Rating + Disagreement + Artist.Male + Artist.White +
##   Artist.Black + Vocally.Based + Bass.Based + Guitar.Based +
##   Saxophone + Trumpet + Written.for.a.Film + Decade
## Model 2: Long.Stay ~ BPM + Energy + Danceability + Happiness + Acousticness +
##   Length + Overall.Rating + Disagreement + Artist.Male + Artist.White +
##   Artist.Black + Vocally.Based + Guitar.Based + Saxophone +
##   Trumpet + Written.for.a.Film + Decade
##   Resid. Df Resid. Dev Df   Deviance Pr(>Chi)
## 1      1153      1275.6
## 2      1154      1275.6 -1 -0.0011363   0.9731

summary(m3)

##
## Call:
## glm(formula = Long.Stay ~ BPM + Energy + Danceability + Happiness +
##   Acousticness + Length + Overall.Rating + Disagreement + Artist.Male +
##   Artist.White + Artist.Black + Vocally.Based + Guitar.Based +

```

```
##      Saxophone + Trumpet + Written.for.a.Film + Decade, family = binomial,
##      data = billboard)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)   -3.650e+01  9.354e+00  -3.902 9.54e-05 ***
## BPM           -7.897e-04  2.862e-03  -0.276 0.782607
## Energy        -9.232e-03  4.934e-03  -1.871 0.061309 .
## Danceability    6.289e-03  5.957e-03   1.056 0.291092
## Happiness      1.145e-03  3.902e-03   0.294 0.769132
## Acousticness   -1.886e-03  3.409e-03  -0.553 0.580166
## Length         4.137e-04  1.405e-03   0.295 0.768346
## Overall.Rating  2.073e-01  4.211e-02   4.924 8.50e-07 ***
## Disagreement   1.249e-01  6.406e-02   1.950 0.051219 .
## Artist.Male    1.384e-01  1.111e-01   1.245 0.213119
## Artist.White   1.500e-01  1.119e-01   1.340 0.180105
## Artist.Black   1.039e-01  1.110e-01   0.936 0.349382
## Vocally.Based  1.461e-01  3.239e-01   0.451 0.651969
## Guitar.Based   -3.570e-01  1.521e-01  -2.348 0.018890 *
## Saxophone     -1.328e-01  3.334e-01  -0.398 0.690343
## Trumpet        -6.998e-01  6.487e-01  -1.079 0.280638
## Written.for.a.Film 1.962e-01  2.486e-01   0.789 0.430086
## Decade         1.713e-02  4.719e-03   3.631 0.000282 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 1350.1  on 1171  degrees of freedom
## Residual deviance: 1275.6  on 1154  degrees of freedom
## AIC: 1311.6
##
## Number of Fisher Scoring iterations: 4
m4 <- update(m3, . ~ . - Artist.Male)
anova(m3, m4, test = "LRT")

## Analysis of Deviance Table
##
## Model 1: Long.Stay ~ BPM + Energy + Danceability + Happiness + Acousticness +
##      Length + Overall.Rating + Disagreement + Artist.Male + Artist.White +
##      Artist.Black + Vocally.Based + Guitar.Based + Saxophone +
##      Trumpet + Written.for.a.Film + Decade
## Model 2: Long.Stay ~ BPM + Energy + Danceability + Happiness + Acousticness +
##      Length + Overall.Rating + Disagreement + Artist.White + Artist.Black +
##      Vocally.Based + Guitar.Based + Saxophone + Trumpet + Written.for.a.Film +
##      Decade
##      Resid. Df Resid. Dev Df Deviance Pr(>Chi)
## 1          1154      1275.6
## 2          1155      1277.1 -1   -1.5511   0.213

summary(m4)

##
## Call:
```

```
## glm(formula = Long.Stay ~ BPM + Energy + Danceability + Happiness +
##       Acousticness + Length + Overall.Rating + Disagreement + Artist.White +
##       Artist.Black + Vocally.Based + Guitar.Based + Saxophone +
##       Trumpet + Written.for.a.Film + Decade, family = binomial,
##       data = billboard)
##
## Coefficients:
##               Estimate Std. Error z value Pr(>|z|)
## (Intercept)   -3.597e+01  9.342e+00 -3.851 0.000118 ***
## BPM           -8.216e-04  2.861e-03  -0.287 0.773938
## Energy        -9.654e-03  4.921e-03  -1.962 0.049761 *
## Danceability    5.942e-03  5.958e-03   0.997 0.318630
## Happiness      1.467e-03  3.891e-03   0.377 0.706196
## Acousticness  -2.000e-03  3.405e-03  -0.587 0.557002
## Length         4.390e-04  1.406e-03   0.312 0.754850
## Overall.Rating  2.018e-01  4.185e-02   4.823 1.41e-06 ***
## Disagreement   1.261e-01  6.402e-02   1.969 0.048947 *
## Artist.White   1.728e-01  1.103e-01   1.567 0.117128
## Artist.Black   1.320e-01  1.085e-01   1.217 0.223697
## Vocally.Based  1.472e-01  3.236e-01   0.455 0.649209
## Guitar.Based  -3.486e-01  1.518e-01  -2.297 0.021631 *
## Saxophone     -1.463e-01  3.333e-01  -0.439 0.660631
## Trumpet       -6.557e-01  6.469e-01  -1.014 0.310772
## Written.for.a.Film 1.964e-01  2.484e-01   0.791 0.429127
## Decade        1.694e-02  4.716e-03   3.593 0.000327 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 1350.1  on 1171  degrees of freedom
## Residual deviance: 1277.1  on 1155  degrees of freedom
## AIC: 1311.1
##
## Number of Fisher Scoring iterations: 4
m5 <- update(m4, . ~ . - BPM)
anova(m4, m5, test = "LRT")

## Analysis of Deviance Table
##
## Model 1: Long.Stay ~ BPM + Energy + Danceability + Happiness + Acousticness +
##       Length + Overall.Rating + Disagreement + Artist.White + Artist.Black +
##       Vocally.Based + Guitar.Based + Saxophone + Trumpet + Written.for.a.Film +
##       Decade
## Model 2: Long.Stay ~ Energy + Danceability + Happiness + Acousticness +
##       Length + Overall.Rating + Disagreement + Artist.White + Artist.Black +
##       Vocally.Based + Guitar.Based + Saxophone + Trumpet + Written.for.a.Film +
##       Decade
##   Resid. Df Resid. Dev Df  Deviance Pr(>Chi)
## 1      1155      1277.1
## 2      1156      1277.2 -1 -0.082595  0.7738
```

```
summary(m5)
```

```
##
## Call:
## glm(formula = Long.Stay ~ Energy + Danceability + Happiness +
##      Acousticness + Length + Overall.Rating + Disagreement + Artist.White +
##      Artist.Black + Vocally.Based + Guitar.Based + Saxophone +
##      Trumpet + Written.for.a.Film + Decade, family = binomial,
##      data = billboard)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)   -3.596e+01  9.342e+00  -3.850 0.000118 ***
## Energy        -9.807e-03  4.892e-03  -2.005 0.045004 *
## Danceability    6.081e-03  5.941e-03   1.024 0.306010
## Happiness      1.483e-03  3.891e-03   0.381 0.703134
## Acousticness   -1.939e-03  3.398e-03  -0.571 0.568298
## Length         4.579e-04  1.404e-03   0.326 0.744309
## Overall.Rating  2.020e-01  4.185e-02   4.826 1.39e-06 ***
## Disagreement   1.265e-01  6.403e-02   1.975 0.048232 *
## Artist.White   1.723e-01  1.103e-01   1.563 0.118150
## Artist.Black   1.335e-01  1.084e-01   1.232 0.217764
## Vocally.Based  1.469e-01  3.235e-01   0.454 0.649727
## Guitar.Based   -3.485e-01  1.518e-01  -2.296 0.021690 *
## Saxophone      -1.455e-01  3.331e-01  -0.437 0.662261
## Trumpet        -6.599e-01  6.469e-01  -1.020 0.307651
## Written.for.a.Film 1.954e-01  2.484e-01   0.787 0.431517
## Decade         1.689e-02  4.712e-03   3.584 0.000338 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 1350.1  on 1171  degrees of freedom
## Residual deviance: 1277.2  on 1156  degrees of freedom
## AIC: 1309.2
##
## Number of Fisher Scoring iterations: 4
m6 <- update(m5, . ~ . - Length)
anova(m5, m6, test = "LRT")

## Analysis of Deviance Table
##
## Model 1: Long.Stay ~ Energy + Danceability + Happiness + Acousticness +
##      Length + Overall.Rating + Disagreement + Artist.White + Artist.Black +
##      Vocally.Based + Guitar.Based + Saxophone + Trumpet + Written.for.a.Film +
##      Decade
## Model 2: Long.Stay ~ Energy + Danceability + Happiness + Acousticness +
##      Overall.Rating + Disagreement + Artist.White + Artist.Black +
##      Vocally.Based + Guitar.Based + Saxophone + Trumpet + Written.for.a.Film +
##      Decade
##      Resid. Df Resid. Dev Df Deviance Pr(>Chi)
## 1          1156      1277.2
```

```
## 2      1157      1277.3 -1 -0.1058    0.745
```

```
summary(m6)
```

```
##
## Call:
## glm(formula = Long.Stay ~ Energy + Danceability + Happiness +
##      Acousticness + Overall.Rating + Disagreement + Artist.White +
##      Artist.Black + Vocally.Based + Guitar.Based + Saxophone +
##      Trumpet + Written.for.a.Film + Decade, family = binomial,
##      data = billboard)
##
```

```
## Coefficients:
```

	Estimate	Std. Error	z value	Pr(> z)	
## (Intercept)	-35.907745	9.330329	-3.848	0.000119	***
## Energy	-0.009785	0.004893	-2.000	0.045498	*
## Danceability	0.006225	0.005923	1.051	0.293314	
## Happiness	0.001218	0.003805	0.320	0.748930	
## Acousticness	-0.002087	0.003368	-0.620	0.535578	
## Overall.Rating	0.202621	0.041791	4.848	1.24e-06	***
## Disagreement	0.125394	0.063939	1.961	0.049862	*
## Artist.White	0.172877	0.110268	1.568	0.116930	
## Artist.Black	0.135169	0.108234	1.249	0.211716	
## Vocally.Based	0.134247	0.320935	0.418	0.675728	
## Guitar.Based	-0.351351	0.151557	-2.318	0.020435	*
## Saxophone	-0.144707	0.333029	-0.435	0.663912	
## Trumpet	-0.652559	0.646353	-1.010	0.312687	
## Written.for.a.Film	0.198742	0.248133	0.801	0.423162	
## Decade	0.016916	0.004706	3.595	0.000325	***

```
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
```

```
## (Dispersion parameter for binomial family taken to be 1)
```

```
##
```

```
## Null deviance: 1350.1 on 1171 degrees of freedom
```

```
## Residual deviance: 1277.3 on 1157 degrees of freedom
```

```
## AIC: 1307.3
```

```
##
```

```
## Number of Fisher Scoring iterations: 4
```

```
m7 <- update(m6, . ~ . - Happiness)
```

```
anova(m6, m7, test = "LRT")
```

```
## Analysis of Deviance Table
```

```
##
```

```
## Model 1: Long.Stay ~ Energy + Danceability + Happiness + Acousticness +
##      Overall.Rating + Disagreement + Artist.White + Artist.Black +
##      Vocally.Based + Guitar.Based + Saxophone + Trumpet + Written.for.a.Film +
##      Decade
```

```
## Model 2: Long.Stay ~ Energy + Danceability + Acousticness + Overall.Rating +
##      Disagreement + Artist.White + Artist.Black + Vocally.Based +
##      Guitar.Based + Saxophone + Trumpet + Written.for.a.Film +
##      Decade
```

```
## Resid. Df Resid. Dev Df Deviance Pr(>Chi)
```

```
## 1      1157      1277.3
```

```
## 2      1158      1277.4 -1 -0.10254    0.7488
```

```
summary(m7)
```

```
##
## Call:
## glm(formula = Long.Stay ~ Energy + Danceability + Acousticness +
##      Overall.Rating + Disagreement + Artist.White + Artist.Black +
##      Vocally.Based + Guitar.Based + Saxophone + Trumpet + Written.for.a.Film +
##      Decade, family = binomial, data = billboard)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)   -34.760640    8.616413  -4.034 5.48e-05 ***
## Energy         -0.009098    0.004393  -2.071 0.038362 *
## Danceability    0.007096    0.005262   1.349 0.177492
## Acousticness   -0.002031    0.003363  -0.604 0.545896
## Overall.Rating  0.202981    0.041779   4.858 1.18e-06 ***
## Disagreement   0.125368    0.063950   1.960 0.049949 *
## Artist.White   0.175045    0.110069   1.590 0.111762
## Artist.Black   0.136022    0.108219   1.257 0.208784
## Vocally.Based  0.133082    0.320860   0.415 0.678313
## Guitar.Based   -0.346072    0.150652  -2.297 0.021610 *
## Saxophone      -0.138194    0.332300  -0.416 0.677505
## Trumpet        -0.645061    0.645759  -0.999 0.317834
## Written.for.a.Film 0.194235    0.247602   0.784 0.432767
## Decade         0.016322    0.004326   3.773 0.000161 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 1350.1  on 1171  degrees of freedom
## Residual deviance: 1277.4  on 1158  degrees of freedom
## AIC: 1305.4
##
## Number of Fisher Scoring iterations: 4
```

```
m8 <- update(m7, . ~ . - Vocally.Based)
anova(m7, m8, test = "LRT")
```

```
## Analysis of Deviance Table
##
## Model 1: Long.Stay ~ Energy + Danceability + Acousticness + Overall.Rating +
##      Disagreement + Artist.White + Artist.Black + Vocally.Based +
##      Guitar.Based + Saxophone + Trumpet + Written.for.a.Film +
##      Decade
## Model 2: Long.Stay ~ Energy + Danceability + Acousticness + Overall.Rating +
##      Disagreement + Artist.White + Artist.Black + Guitar.Based +
##      Saxophone + Trumpet + Written.for.a.Film + Decade
##      Resid. Df Resid. Dev Df Deviance Pr(>Chi)
## 1      1158      1277.4
## 2      1159      1277.6 -1   -0.1692    0.6808
```

```
summary(m8)
```

```
##
## Call:
## glm(formula = Long.Stay ~ Energy + Danceability + Acousticness +
##      Overall.Rating + Disagreement + Artist.White + Artist.Black +
##      Guitar.Based + Saxophone + Trumpet + Written.for.a.Film +
##      Decade, family = binomial, data = billboard)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)   -34.213896    8.513868  -4.019 5.85e-05 ***
## Energy         -0.009127    0.004395  -2.077 0.037823 *
## Danceability    0.007067    0.005260   1.343 0.179142
## Acousticness   -0.001877    0.003344  -0.561 0.574598
## Overall.Rating  0.203156    0.041751   4.866 1.14e-06 ***
## Disagreement   0.124786    0.063929   1.952 0.050944 .
## Artist.White   0.173640    0.110013   1.578 0.114484
## Artist.Black   0.136793    0.108187   1.264 0.206082
## Guitar.Based   -0.348368    0.150549  -2.314 0.020669 *
## Saxophone      -0.141040    0.332236  -0.425 0.671189
## Trumpet        -0.653759    0.645213  -1.013 0.310942
## Written.for.a.Film 0.191016    0.247359   0.772 0.439983
## Decade         0.016051    0.004276   3.754 0.000174 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 1350.1  on 1171  degrees of freedom
## Residual deviance: 1277.6  on 1159  degrees of freedom
## AIC: 1303.6
##
## Number of Fisher Scoring iterations: 4
m9 <- update(m8, . ~ . - Saxophone)
anova(m8, m9, test = "LRT")

## Analysis of Deviance Table
##
## Model 1: Long.Stay ~ Energy + Danceability + Acousticness + Overall.Rating +
##      Disagreement + Artist.White + Artist.Black + Guitar.Based +
##      Saxophone + Trumpet + Written.for.a.Film + Decade
## Model 2: Long.Stay ~ Energy + Danceability + Acousticness + Overall.Rating +
##      Disagreement + Artist.White + Artist.Black + Guitar.Based +
##      Trumpet + Written.for.a.Film + Decade
##   Resid. Df Resid. Dev Df Deviance Pr(>Chi)
## 1      1159      1277.6
## 2      1160      1277.8 -1 -0.18406   0.6679

summary(m9)

##
## Call:
## glm(formula = Long.Stay ~ Energy + Danceability + Acousticness +
##      Overall.Rating + Disagreement + Artist.White + Artist.Black +
##      Guitar.Based + Trumpet + Written.for.a.Film + Decade, family = binomial,
```

```
## data = billboard)
##
## Coefficients:
##           Estimate Std. Error z value Pr(>|z|)
## (Intercept)   -34.544683    8.478557  -4.074 4.61e-05 ***
## Energy         -0.009056    0.004389  -2.063 0.039098 *
## Danceability    0.007066    0.005259   1.344 0.179032
## Acousticness   -0.001815    0.003339  -0.544 0.586763
## Overall.Rating  0.203401    0.041735   4.874 1.10e-06 ***
## Disagreement   0.125184    0.063933   1.958 0.050224 .
## Artist.White   0.171758    0.109933   1.562 0.118197
## Artist.Black   0.136067    0.108124   1.258 0.208232
## Guitar.Based   -0.348302    0.150518  -2.314 0.020667 *
## Trumpet        -0.667414    0.645089  -1.035 0.300852
## Written.for.a.Film 0.192071    0.247356   0.776 0.437456
## Decade         0.016211    0.004259   3.806 0.000141 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
## Null deviance: 1350.1 on 1171 degrees of freedom
## Residual deviance: 1277.8 on 1160 degrees of freedom
## AIC: 1301.8
##
## Number of Fisher Scoring iterations: 4
m10 <- update(m9, . ~ . - Acousticness)
anova(m9, m10, test = "LRT")

## Analysis of Deviance Table
##
## Model 1: Long.Stay ~ Energy + Danceability + Acousticness + Overall.Rating +
## Disagreement + Artist.White + Artist.Black + Guitar.Based +
## Trumpet + Written.for.a.Film + Decade
## Model 2: Long.Stay ~ Energy + Danceability + Overall.Rating + Disagreement +
## Artist.White + Artist.Black + Guitar.Based + Trumpet + Written.for.a.Film +
## Decade
## Resid. Df Resid. Dev Df Deviance Pr(>Chi)
## 1      1160      1277.8
## 2      1161      1278.1 -1 -0.29599  0.5864

summary(m10)

##
## Call:
## glm(formula = Long.Stay ~ Energy + Danceability + Overall.Rating +
## Disagreement + Artist.White + Artist.Black + Guitar.Based +
## Trumpet + Written.for.a.Film + Decade, family = binomial,
## data = billboard)
##
## Coefficients:
##           Estimate Std. Error z value Pr(>|z|)
## (Intercept)   -35.790500    8.155776  -4.388 1.14e-05 ***
## Energy         -0.007944    0.003878  -2.048  0.0405 *
```

```
## Danceability      0.007798  0.005088  1.533  0.1253
## Overall.Rating    0.203569  0.041719  4.880 1.06e-06 ***
## Disagreement      0.124968  0.063907  1.955  0.0505 .
## Artist.White      0.172469  0.109928  1.569  0.1167
## Artist.Black      0.136232  0.108151  1.260  0.2078
## Guitar.Based     -0.345673  0.150366 -2.299  0.0215 *
## Trumpet          -0.674452  0.644980 -1.046  0.2957
## Written.for.a.Film 0.201681  0.246836  0.817  0.4139
## Decade            0.016754  0.004136  4.050 5.11e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
## Null deviance: 1350.1 on 1171 degrees of freedom
## Residual deviance: 1278.1 on 1161 degrees of freedom
## AIC: 1300.1
##
## Number of Fisher Scoring iterations: 4
```

```
m11 <- update(m10, . ~ . - Written.for.a.Film)
anova(m10, m11, test = "LRT")
```

```
## Analysis of Deviance Table
##
## Model 1: Long.Stay ~ Energy + Danceability + Overall.Rating + Disagreement +
## Artist.White + Artist.Black + Guitar.Based + Trumpet + Written.for.a.Film +
## Decade
## Model 2: Long.Stay ~ Energy + Danceability + Overall.Rating + Disagreement +
## Artist.White + Artist.Black + Guitar.Based + Trumpet + Decade
## Resid. Df Resid. Dev Df Deviance Pr(>Chi)
## 1      1161      1278.1
## 2      1162      1278.7 -1  -0.6536  0.4188
```

```
summary(m11)
```

```
##
## Call:
## glm(formula = Long.Stay ~ Energy + Danceability + Overall.Rating +
## Disagreement + Artist.White + Artist.Black + Guitar.Based +
## Trumpet + Decade, family = binomial, data = billboard)
##
## Coefficients:
## Estimate Std. Error z value Pr(>|z|)
## (Intercept) -35.536806  8.136472 -4.368 1.26e-05 ***
## Energy      -0.008110  0.003870 -2.096  0.0361 *
## Danceability  0.007638  0.005079  1.504  0.1327
## Overall.Rating 0.203174  0.041673  4.875 1.09e-06 ***
## Disagreement  0.125799  0.063889  1.969  0.0489 *
## Artist.White  0.172626  0.109917  1.571  0.1163
## Artist.Black  0.135174  0.108110  1.250  0.2112
## Guitar.Based -0.349699  0.150341 -2.326  0.0200 *
## Trumpet      -0.691199  0.644501 -1.072  0.2835
## Decade       0.016646  0.004128  4.033 5.51e-05 ***
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##    Null deviance: 1350.1  on 1171  degrees of freedom
## Residual deviance: 1278.7  on 1162  degrees of freedom
## AIC: 1298.7
##
## Number of Fisher Scoring iterations: 4
m12 <- update(m11, . ~ . - Trumpet)
anova(m11, m12, test = "LRT")

## Analysis of Deviance Table
##
## Model 1: Long.Stay ~ Energy + Danceability + Overall.Rating + Disagreement +
##   Artist.White + Artist.Black + Guitar.Based + Trumpet + Decade
## Model 2: Long.Stay ~ Energy + Danceability + Overall.Rating + Disagreement +
##   Artist.White + Artist.Black + Guitar.Based + Decade
##   Resid. Df Resid. Dev Df Deviance Pr(>Chi)
## 1      1162      1278.7
## 2      1163      1280.0 -1  -1.3173  0.2511

summary(m12)

##
## Call:
## glm(formula = Long.Stay ~ Energy + Danceability + Overall.Rating +
##   Disagreement + Artist.White + Artist.Black + Guitar.Based +
##   Decade, family = binomial, data = billboard)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)  -35.723001   8.128688  -4.395 1.11e-05 ***
## Energy        -0.007989   0.003870  -2.064  0.0390 *
## Danceability   0.007454   0.005075   1.469  0.1419
## Overall.Rating  0.202404   0.041654   4.859 1.18e-06 ***
## Disagreement   0.125507   0.063879   1.965  0.0494 *
## Artist.White   0.175882   0.109906   1.600  0.1095
## Artist.Black   0.134817   0.108024   1.248  0.2120
## Guitar.Based  -0.350216   0.150225  -2.331  0.0197 *
## Decade         0.016739   0.004124   4.059 4.93e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##    Null deviance: 1350.1  on 1171  degrees of freedom
## Residual deviance: 1280.0  on 1163  degrees of freedom
## AIC: 1298
##
## Number of Fisher Scoring iterations: 4
m13 <- update(m12, . ~ . - Artist.Black)
anova(m12, m13, test = "LRT")
```

```
## Analysis of Deviance Table
##
## Model 1: Long.Stay ~ Energy + Danceability + Overall.Rating + Disagreement +
##   Artist.White + Artist.Black + Guitar.Based + Decade
## Model 2: Long.Stay ~ Energy + Danceability + Overall.Rating + Disagreement +
##   Artist.White + Guitar.Based + Decade
##   Resid. Df Resid. Dev Df Deviance Pr(>Chi)
## 1      1163      1280.0
## 2      1164      1281.6 -1  -1.5455  0.2138
```

```
summary(m13)
```

```
##
## Call:
## glm(formula = Long.Stay ~ Energy + Danceability + Overall.Rating +
##   Disagreement + Artist.White + Guitar.Based + Decade, family = binomial,
##   data = billboard)
##
## Coefficients:
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept)  -37.103201   8.042583  -4.613 3.96e-06 ***
## Energy        -0.007962   0.003865  -2.060  0.0394 *
## Danceability   0.008725   0.004975   1.754  0.0795 .
## Overall.Rating  0.205427   0.041570   4.942 7.74e-07 ***
## Disagreement   0.128245   0.063764   2.011  0.0443 *
## Artist.White    0.171652   0.111200   1.544  0.1227
## Guitar.Based   -0.369572   0.149237  -2.476  0.0133 *
## Decade         0.017425   0.004082   4.269 1.97e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##   Null deviance: 1350.1  on 1171  degrees of freedom
## Residual deviance: 1281.6  on 1164  degrees of freedom
## AIC: 1297.6
##
## Number of Fisher Scoring iterations: 4
```

```
m14 <- update(m13, . ~ . - Artist.White)
anova(m13, m14, test = "LRT")
```

```
## Analysis of Deviance Table
##
## Model 1: Long.Stay ~ Energy + Danceability + Overall.Rating + Disagreement +
##   Artist.White + Guitar.Based + Decade
## Model 2: Long.Stay ~ Energy + Danceability + Overall.Rating + Disagreement +
##   Guitar.Based + Decade
##   Resid. Df Resid. Dev Df Deviance Pr(>Chi)
## 1      1164      1281.6
## 2      1165      1284.0 -1  -2.3806  0.1228
```

```
summary(m14)
```

```
##
## Call:
```

```
## glm(formula = Long.Stay ~ Energy + Danceability + Overall.Rating +
##       Disagreement + Guitar.Based + Decade, family = binomial,
##       data = billboard)
##
## Coefficients:
##               Estimate Std. Error z value Pr(>|z|)
## (Intercept)  -36.683659   8.033753  -4.566 4.97e-06 ***
## Energy        -0.007542   0.003852  -1.958  0.0502 .
## Danceability   0.007889   0.004942   1.596  0.1104
## Overall.Rating  0.194315   0.040877   4.754 2.00e-06 ***
## Disagreement   0.125439   0.063603   1.972  0.0486 *
## Guitar.Based  -0.333133   0.147219  -2.263  0.0236 *
## Decade         0.017317   0.004079   4.245 2.19e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 1350.1  on 1171  degrees of freedom
## Residual deviance: 1284.0  on 1165  degrees of freedom
## AIC: 1298
##
## Number of Fisher Scoring iterations: 4
m15 <- update(m14, . ~ . - Danceability)
anova(m14, m15, test = "LRT")

## Analysis of Deviance Table
##
## Model 1: Long.Stay ~ Energy + Danceability + Overall.Rating + Disagreement +
##       Guitar.Based + Decade
## Model 2: Long.Stay ~ Energy + Overall.Rating + Disagreement + Guitar.Based +
##       Decade
##      Resid. Df Resid. Dev Df Deviance Pr(>Chi)
## 1          1165      1284.0
## 2          1166      1286.5 -1    -2.563   0.1094

summary(m15)

##
## Call:
## glm(formula = Long.Stay ~ Energy + Overall.Rating + Disagreement +
##       Guitar.Based + Decade, family = binomial, data = billboard)
##
## Coefficients:
##               Estimate Std. Error z value Pr(>|z|)
## (Intercept)  -39.653375   7.803177  -5.082 3.74e-07 ***
## Energy        -0.006005   0.003711  -1.618  0.1056
## Overall.Rating  0.197848   0.040868   4.841 1.29e-06 ***
## Disagreement   0.126506   0.063589   1.989  0.0467 *
## Guitar.Based  -0.332699   0.146898  -2.265  0.0235 *
## Decade         0.019007   0.003933   4.832 1.35e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
```

```
## (Dispersion parameter for binomial family taken to be 1)
##
## Null deviance: 1350.1 on 1171 degrees of freedom
## Residual deviance: 1286.5 on 1166 degrees of freedom
## AIC: 1298.5
##
## Number of Fisher Scoring iterations: 4
m16 <- update(m15, . ~ . - Energy)
anova(m15, m16, test = "LRT")

## Analysis of Deviance Table
##
## Model 1: Long.Stay ~ Energy + Overall.Rating + Disagreement + Guitar.Based +
## Decade
## Model 2: Long.Stay ~ Overall.Rating + Disagreement + Guitar.Based + Decade
## Resid. Df Resid. Dev Df Deviance Pr(>Chi)
## 1 1166 1286.5
## 2 1167 1289.2 -1 -2.6215 0.1054

summary(m16)

##
## Call:
## glm(formula = Long.Stay ~ Overall.Rating + Disagreement + Guitar.Based +
## Decade, family = binomial, data = billboard)
##
## Coefficients:
## Estimate Std. Error z value Pr(>|z|)
## (Intercept) -37.04916 7.62758 -4.857 1.19e-06 ***
## Overall.Rating 0.19412 0.04073 4.766 1.88e-06 ***
## Disagreement 0.12611 0.06355 1.984 0.0472 *
## Guitar.Based -0.34413 0.14659 -2.348 0.0189 *
## Decade 0.01753 0.00382 4.588 4.48e-06 ***
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
## Null deviance: 1350.1 on 1171 degrees of freedom
## Residual deviance: 1289.2 on 1167 degrees of freedom
## AIC: 1299.2
##
## Number of Fisher Scoring iterations: 4
m17 <- update(m16, . ~ . - Disagreement)
anova(m16, m17, test = "LRT")

## Analysis of Deviance Table
##
## Model 1: Long.Stay ~ Overall.Rating + Disagreement + Guitar.Based + Decade
## Model 2: Long.Stay ~ Overall.Rating + Guitar.Based + Decade
## Resid. Df Resid. Dev Df Deviance Pr(>Chi)
## 1 1167 1289.2
## 2 1168 1293.0 -1 -3.8438 0.04993 *
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
summary(m13)
```

```
##
## Call:
## glm(formula = Long.Stay ~ Energy + Danceability + Overall.Rating +
##      Disagreement + Artist.White + Guitar.Based + Decade, family = binomial,
##      data = billboard)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)  -37.103201   8.042583  -4.613 3.96e-06 ***
## Energy        -0.007962   0.003865  -2.060  0.0394 *
## Danceability   0.008725   0.004975   1.754  0.0795 .
## Overall.Rating  0.205427   0.041570   4.942 7.74e-07 ***
## Disagreement   0.128245   0.063764   2.011  0.0443 *
## Artist.White    0.171652   0.111200   1.544  0.1227
## Guitar.Based   -0.369572   0.149237  -2.476  0.0133 *
## Decade          0.017425   0.004082   4.269 1.97e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 1350.1  on 1171  degrees of freedom
## Residual deviance: 1281.6  on 1164  degrees of freedom
## AIC: 1297.6
##
## Number of Fisher Scoring iterations: 4
```

Let's try a few interactions: Disagreement * Overall.Rating, Energy * Danceability

```
m13.1 <- update(m13, . ~ . + Overall.Rating:Disagreement)
m13.2 <- update(m13, . ~ . + Energy:Danceability)
m13.3 <- update(m13, . ~ . + Energy:Danceability + Overall.Rating:Disagreement)
m13.4 <- update(m13, . ~ . + Guitar.Based:Decade)
m13.5 <- update(m13, . ~ . + Danceability:Artist.White + Overall.Rating:Disagreement)
m13.6 <- update(m13, . ~ . + Danceability:Artist.White + Energy:Decade)
```

```
anova(m13, m13.6, test = "LRT")
```

```
## Analysis of Deviance Table
##
## Model 1: Long.Stay ~ Energy + Danceability + Overall.Rating + Disagreement +
##      Artist.White + Guitar.Based + Decade
## Model 2: Long.Stay ~ Energy + Danceability + Overall.Rating + Disagreement +
##      Artist.White + Guitar.Based + Decade + Danceability:Artist.White +
##      Energy:Decade
##   Resid. Df Resid. Dev Df Deviance Pr(>Chi)
## 1         1164      1281.6
## 2         1162      1271.8  2    9.7609 0.007594 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```

AIC(m13)

## [1] 1297.585
AIC(m13.1)

## [1] 1296.523
AIC(m13.2)

## [1] 1299.209
AIC(m13.3)

## [1] 1298.042
AIC(m13.4)

## [1] 1298.384
AIC(m13.5)

## [1] 1295.236
AIC(m13.6) # Final Model

## [1] 1291.824
library(ggplot2)
library(dplyr)

df <- data # correct dataset

# ---- 1. Generate energy range ----
energy_seq <- seq(
  min(df$Energy, na.rm = TRUE),
  max(df$Energy, na.rm = TRUE),
  length.out = 200
)

# ---- 2. Identify MOST COMMON integer levels for the fixed variables ----
mode_Artist.White <- as.integer(names(sort(table(df$Artist.White), decreasing = TRUE))[1])
mode_Guitar.Based <- as.integer(names(sort(table(df$Guitar.Based), decreasing = TRUE))[1])

# ---- 3. Create prediction grid ----
newdat <- expand.grid(
  Energy = energy_seq,
  Decade = sort(unique(df$Decade)), # integer
  Danceability = mean(df$Danceability, na.rm = TRUE),
  Overall.Rating = mean(df$Overall.Rating, na.rm = TRUE),
  Disagreement = mean(df$Disagreement, na.rm = TRUE),

  Artist.White = mode_Artist.White, # integer, correct type
  Guitar.Based = mode_Guitar.Based # integer, correct type
)

# ---- 4. Predictions ----
pred <- predict(m13.6, newdata = newdat, type = "link", se.fit = TRUE)

```

```

newdat$fit <- plogis(pred$fit)
newdat$lo <- plogis(pred$fit - 1.96 * pred$se.fit)
newdat$hi <- plogis(pred$fit + 1.96 * pred$se.fit)

# ---- 5. Clean Decade labels for plotting ----
newdat$Decade <- factor(newdat$Decade,
                        levels = sort(unique(df$Decade)),
                        labels = paste0(sort(unique(df$Decade)), "s"))

# ---- 6. Plot ----
p <- ggplot(newdat, aes(x = Energy, y = fit, color = Decade)) +
  geom_line(size = 1.1) +
  geom_ribbon(aes(ymin = lo, ymax = hi, fill = Decade), alpha = 0.15, color = NA) +
  labs(
    title = "Predicted Probability of Long Stay",
    subtitle = "Interaction: Energy × Decade",
    x = "Energy (0-100)",
    y = "Predicted Probability",
    color = "Decade",
    fill = "Decade"
  ) +
  theme_classic(base_size = 15) +
  theme(
    legend.position = "right",
    plot.title = element_text(face = "bold")
  )

```

```

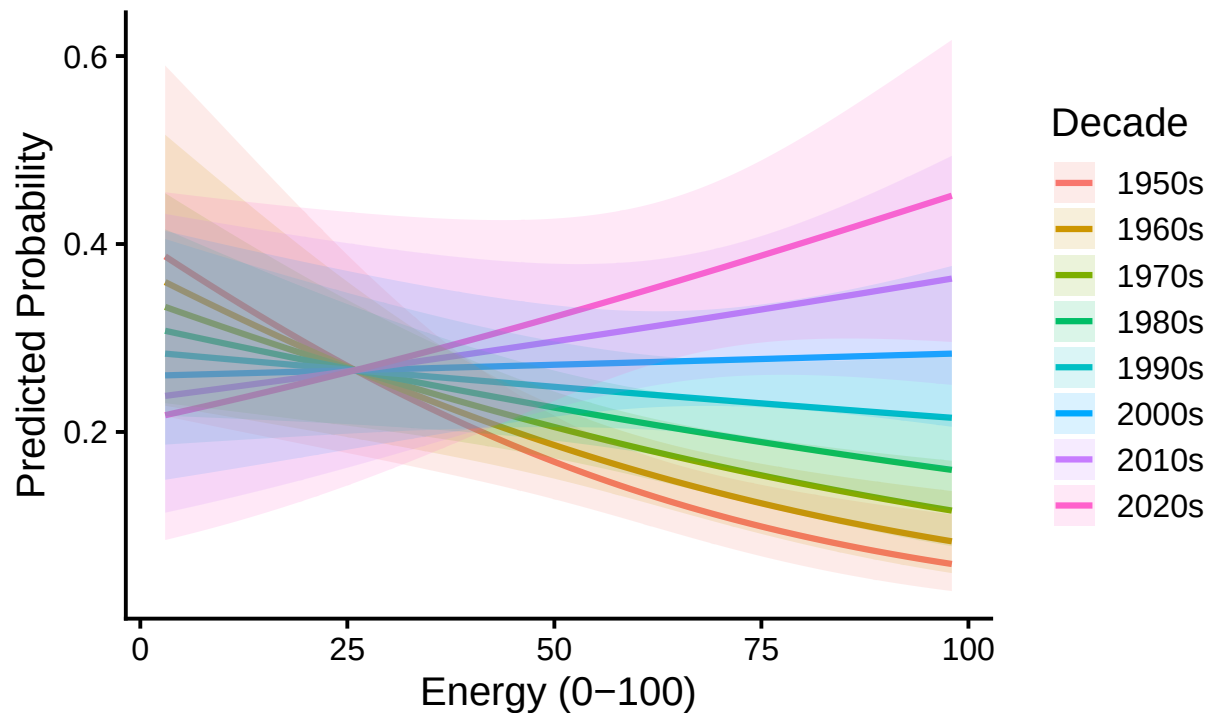
## Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use `linewidth` instead.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.

```

```
p
```

Predicted Probability of Long Stay

Interaction: Energy × Decade



The plot reveals a clear decade-by-decade shift in how song Energy predicts chart longevity. In the 1950s–1980s, higher-energy songs were less likely to remain at #1 for 4+ weeks. By contrast, in recent decades — especially the 2010s and 2020s — higher-energy songs show a higher probability of extended chart dominance. The 1990s–2000s show relatively flat slopes, indicating that Energy was not strongly related to chart longevity during that period. These patterns suggest that the commercial appeal of high-energy tracks has increased sharply over time, reflecting shifts in popular music production, consumption habits, and genre dominance.

```
library(ggplot2)
library(dplyr)

df <- billboard # correct dataset

# ---- 1. Define range of Danceability ----
dance_seq <- seq(
  min(df$Danceability, na.rm = TRUE),
  max(df$Danceability, na.rm = TRUE),
  length.out = 200
)

# ---- 2. Identify most common reference values for non-focal variables ----
mode_Guitar.Based <- as.integer(names(sort(table(df$Guitar.Based), decreasing = TRUE))[1])
mode_Artist.White <- sort(unique(df$Artist.White)) # we will loop over them

# ---- 3. Prediction grid ----
newdat <- expand_grid(
  Danceability = dance_seq,
  Artist.White = sort(unique(df$Artist.White)), # 0,1,2
```

```

Energy          = mean(df$Energy, na.rm = TRUE),
Overall.Rating  = mean(df$Overall.Rating, na.rm = TRUE),
Disagreement    = mean(df$Disagreement, na.rm = TRUE),
Guitar.Based    = mode_Guitar.Based,
Decade          = sort(unique(df$Decade))[1]    # fix Decade
)

# ---- 4. Predictions ----
pred <- predict(m13.6, newdata = newdat, type = "link", se.fit = TRUE)

newdat$fit <- plogis(pred$fit)
newdat$lo  <- plogis(pred$fit - 1.96 * pred$se.fit)
newdat$hi  <- plogis(pred$fit + 1.96 * pred$se.fit)

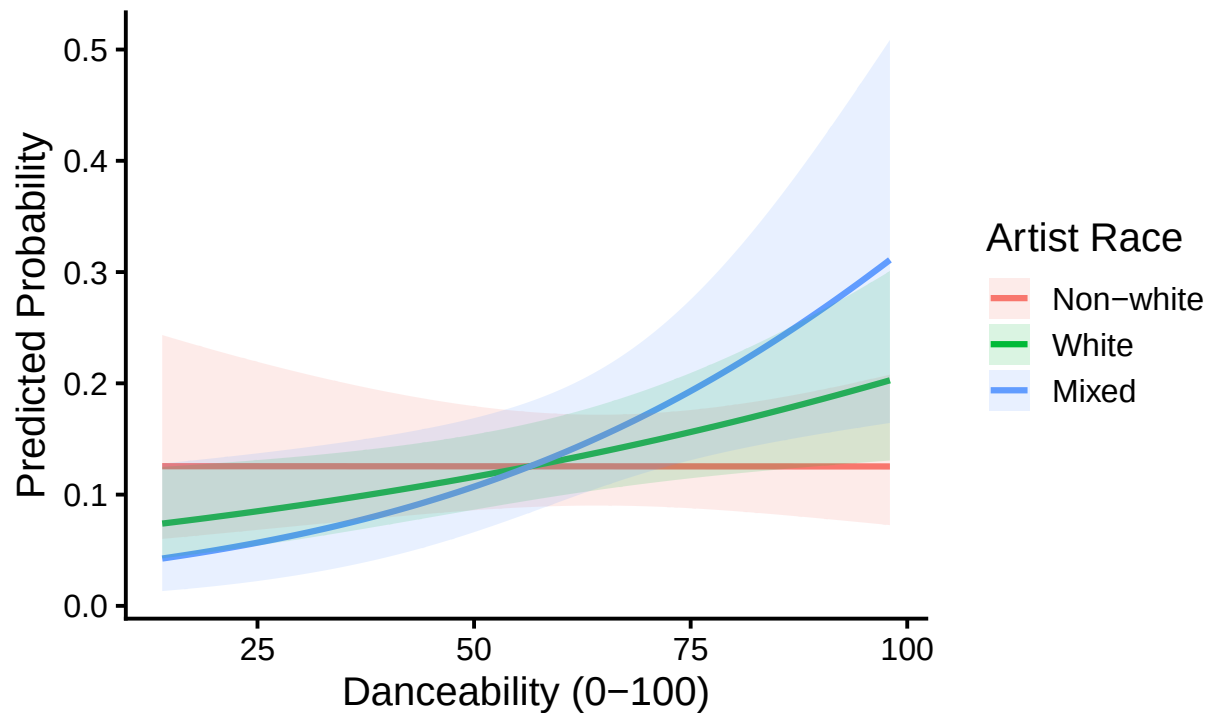
# ---- 5. Clean labels ----
newdat$Artist.White <- factor(
  newdat$Artist.White,
  levels = sort(unique(df$Artist.White)),
  labels = c("Non-white", "White", "Mixed")
)

# ---- 6. Plot ----
p2 <- ggplot(newdat, aes(x = Danceability, y = fit, color = Artist.White)) +
  geom_line(size = 1.1) +
  geom_ribbon(aes(ymin = lo, ymax = hi, fill = Artist.White), alpha = 0.15, color = NA) +
  labs(
    title = "Predicted Probability of Long Stay",
    subtitle = "Interaction: Danceability × Artist.White",
    x = "Danceability (0-100)",
    y = "Predicted Probability",
    color = "Artist Race",
    fill = "Artist Race"
  ) +
  theme_classic(base_size = 15) +
  theme(
    plot.title = element_text(face = "bold"),
    legend.position = "right"
  )
p2

```

Predicted Probability of Long Stay

Interaction: Danceability × Artist.White



The interaction between Danceability and Artist.White reveals substantial differences across racial categories. Among mixed-race artists, higher danceability is strongly associated with increased probability of achieving a long stay at #1. White artists show a moderate positive relationship, with danceability contributing to greater chart longevity but to a lesser extent. In contrast, non-white artists display a flat or slightly negative trend, suggesting that danceability has little effect—or even a small decreasing effect—on the likelihood of long chart duration for this group. These findings indicate that the commercial benefits of producing highly danceable songs differ across artist racial categories.

```
glm(formula = Long.Stay ~ Energy + Danceability + Overall.Rating + Disagreement + Artist.White +
Guitar.Based + Decade + Danceability:Artist.White + Energy:Decade, family = binomial, data = billboard)
```

Coefficient Table (Sore)

```
library(dplyr)
library(broom)
library(kableExtra)

# ----- Number formatting -----
format_num <- function(x) {
  sprintf("%.3f", round(as.numeric(x), 3))
}

# ----- P-value formatting -----
format_p <- function(p) {
  if (is.na(p)) return("N/A")
  if (p < 0.0001) return("$\\mathit{p} < 0.0001^{***}$")
  if (p < 0.001) return(paste0("$\\mathit{p} = ", format_num(p), "^{***}$"))
  if (p < 0.01) return(paste0("$\\mathit{p} = ", format_num(p), "^{**}$"))
}
```

```

  if (p < 0.05) return(paste0("$\\mathit{p} = ", format_num(p), "~{*}$"))
  paste0("$\\mathit{p} = ", format_num(p), "$")
}

# ----- Tidy model: estimates + OR + CI -----
coef_tab <- tidy(m13.6) %>%
  mutate(
    OR      = exp(estimate),
    CI_low  = exp(estimate - 1.96 * std.error),
    CI_high = exp(estimate + 1.96 * std.error)
  )

# ===== NEW SAFE LRT FUNCTION (robust, no NA) =====
lrt_p <- function(term) {

  if (term == "(Intercept)") return(NA_real_)

  # Extract full original model formula
  full_form <- formula(m13.6)
  full_terms <- attr(terms(full_form), "term.labels")

  # Build reduced term list
  reduced_terms <- setdiff(full_terms, term)

  # Rebuild reduced model explicitly (safe)
  reduced_form <- as.formula(
    paste("Long.Stay ~", paste(reduced_terms, collapse = " + "))
  )

  # Fit reduced model safely
  reduced_fit <- try(glm(reduced_form, family = binomial, data = billboard),
    silent = TRUE)
  if (inherits(reduced_fit, "try-error")) return(NA_real_)

  # Perform Likelihood Ratio Test
  LR <- try(anova(reduced_fit, m13.6, test = "LRT"), silent = TRUE)
  if (inherits(LR, "try-error")) return(NA_real_)
  if (!("Pr(>Chi)" %in% names(LR))) return(NA_real_)
  if (nrow(LR) < 2) return(NA_real_)

  LR$`Pr(>Chi)`[2]
}

coef_tab$LRT_raw <- sapply(coef_tab$term, lrt_p)

# ----- Pretty predictor labels -----
name_map <- c(
  "(Intercept)"      = "$\\beta_0$, Intercept",
  "Energy"            = "$\\beta_E$, Energy",
  "Danceability"      = "$\\beta_D$, Danceability",
  "Overall.Rating"    = "$\\beta_R$, Overall Rating",
  "Disagreement"      = "$\\beta_G$, Disagreement",
  "Artist.White"      = "$\\beta_W$, Artist: White",

```

```

"Guitar:Based"          = "$\\beta_B$, Guitar-Based",
"Decade"                = "$\\beta_T$, Decade",
"Danceability:Artist.White" = "$\\beta_{D:W}$, Danceability $\\times$ Artist.White",
"Energy:Decade"         = "$\\beta_{E:T}$, Energy $\\times$ Decade"
)

coef_tab$Predictor_Label <- name_map[coef_tab$term]

# ----- Build final table -----
final_tab <- coef_tab %>%
  mutate(
    Estimate = format_num(estimate),
    OddsRatio = ifelse(term == "(Intercept)", "---", format_num(OR)),
    CI = ifelse(
      term == "(Intercept)",
      "---",
      paste0("(", format_num(CI_low), ", ", format_num(CI_high), ")")
    ),
    LRT_p = sapply(LRT_raw, format_p),
    Predictor = Predictor_Label
  ) %>%
  dplyr::select(Predictor, Estimate, OddsRatio, CI, LRT_p)

# ----- LaTeX output -----
final_tab %>%
  kable(
    format = "latex",
    booktabs = TRUE,
    escape = FALSE,
    caption = "Parameter estimates, odds ratios, 95\\% confidence intervals, and likelihood ratio test p-values",
    col.names = c("Predictor", "Estimate", "Odds Ratio", "95\\% CI", "LRT p-value")
  ) %>%
  kable_styling(latex_options = "hold_position") %>%
  row_spec(0, bold = TRUE) %>%
  row_spec(seq(2, nrow(final_tab), 2), background = "#f0f0f0")

```

Table 4: Parameter estimates, odds ratios, 95% confidence intervals, and likelihood ratio test p-values for the final logistic regression model.

Predictor	Estimate	Odds Ratio	95% CI	LRT p-value
β_0 , Intercept	24.132	—	—	N/A
β_E , Energy	-1.016	0.362	(0.167, 0.785)	$p = 0.010^*$
β_D , Danceability	-0.000	1.000	(0.986, 1.014)	$p = 0.998$
β_R , Overall Rating	0.210	1.234	(1.137, 1.340)	$p < 0.0001^{***}$
β_G , Disagreement	0.133	1.142	(1.007, 1.296)	$p = 0.044^*$
β_W , Artist: White	-0.779	0.459	(0.162, 1.300)	$p = 0.143$
β_B , Guitar-Based	-0.351	0.704	(0.525, 0.944)	$p = 0.019^*$
β_T , Decade	-0.013	0.987	(0.963, 1.012)	$p = 0.300$
$\beta_{D:W}$, Danceability $\times$ Artist.White	0.014	1.014	(0.999, 1.029)	$p = 0.071$
$\beta_{E:T}$, Energy $\times$ Decade	0.001	1.001	(1.000, 1.001)	$p = 0.011^*$

```

library(dplyr)
library(kableExtra)
library(ResourceSelection)

## ResourceSelection 0.3-6    2023-06-27

library(pROC)

## Type 'citation("pROC")' for a citation.
##
## Attaching package: 'pROC'
##
## The following objects are masked from 'package:stats':
##
##      cov, smooth, var

df_model <- billboard %>%
  mutate(
    LongStay = as.numeric(Long.Stay),
    pred_prob = predict(m13.6, type = "response"),
    pred_class = ifelse(pred_prob >= 0.5, 1, 0)
  )

billboard$pred_prob <- predict(m13.6, type = "response")

emp_thresh <- mean(billboard$Long.Stay)

pred_class <- ifelse(billboard$pred_prob >= emp_thresh, 1, 0)

class_tab_raw <- table(
  Predicted = pred_class,
  True      = billboard$Long.Stay
)

class_df <- data.frame(
  Predicted = c("No", "Yes"),
  No        = c(class_tab_raw["0", "0"], class_tab_raw["1", "0"]),
  Yes       = c(class_tab_raw["0", "1"], class_tab_raw["1", "1"])
)

class_kable <- kable(
  class_df,
  format = "latex",
  booktabs = TRUE,
  caption = "Classification Table Using Empirical Probability Threshold (p = 0.2628)",
  align = "c"
) %>%
  kable_styling(latex_options = "hold_position", position = "center") %>%
  add_header_above(c(" " = 1, "True" = 2))

print(class_kable)

hl <- hoslem.test(df_model$LongStay, df_model$pred_prob, g = 10)

```

Table 5: Classification Table Using Empirical Probability Threshold ($p = 0.2628$)

Predicted	True	
	No	Yes
No	549	105
Yes	315	203

```
hl_df <- data.frame(
  Statistic = round(hl$statistic, 3),
  df        = hl$parameter,
  p_value_raw = round(hl$p.value, 4)
)

# Format p-values with LaTeX-safe inline math
hl_df$p_value <- dplyr::case_when(
  hl_df$p_value_raw < 0.0001 ~ "\\(\\mathit{p} < 0.0001^{***}\\)",
  hl_df$p_value_raw < 0.001  ~ paste0("\\(\\mathit{p} = ", hl_df$p_value_raw, "^{***}\\)",
  hl_df$p_value_raw < 0.01   ~ paste0("\\(\\mathit{p} = ", hl_df$p_value_raw, "^{**}\\)",
  hl_df$p_value_raw < 0.05   ~ paste0("\\(\\mathit{p} = ", hl_df$p_value_raw, "^{*}\\)",
  TRUE                        ~ paste0("\\(\\mathit{p} = ", hl_df$p_value_raw, "\\)")
)

# Remove raw p-value and rename safely
hl_df <- hl_df %>%
  dplyr::select(
    Statistic,
    df,
    `p-value` = p_value
  )

hl_kable <- kable(
  hl_df,
  format = "latex",
  booktabs = TRUE,
  escape = FALSE,
  caption = "Hosmer--Lemeshow Goodness-of-Fit Test"
) %>%
  kable_styling(latex_options = "hold_position")

print(hl_kable)
```

Table 6: Hosmer–Lemeshow Goodness-of-Fit Test

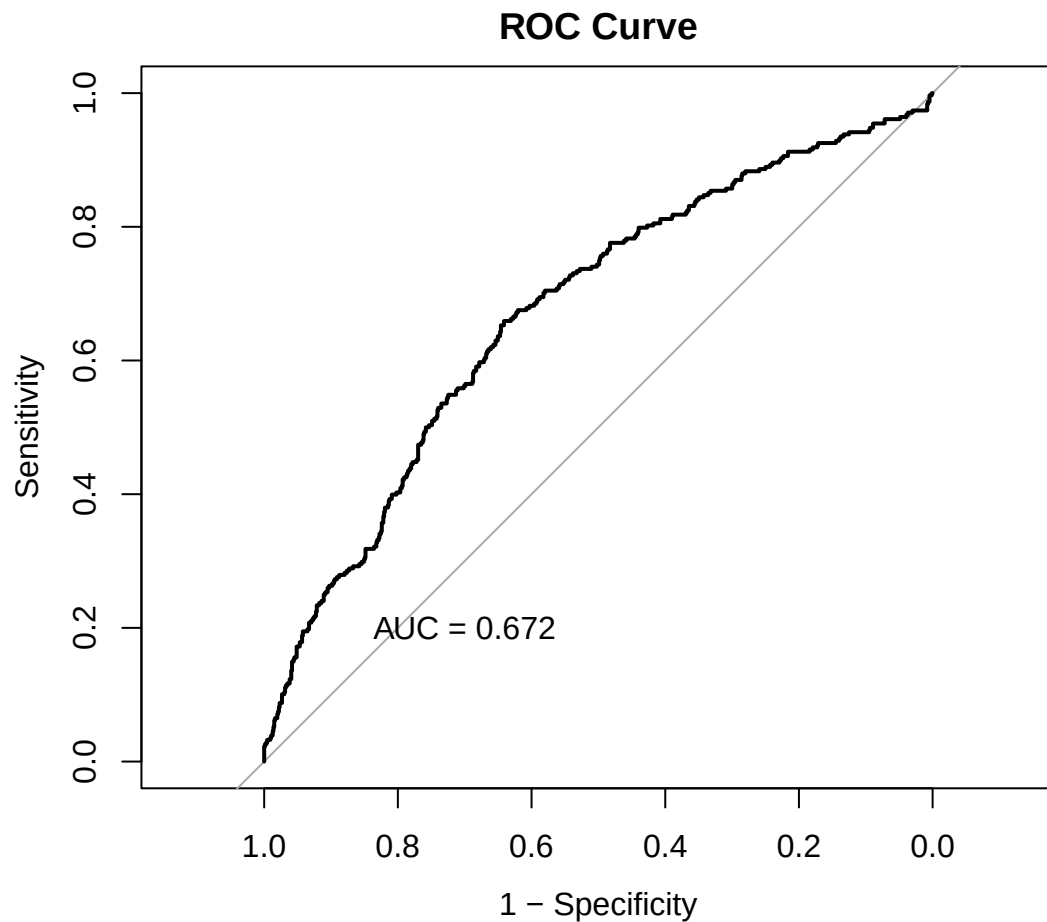
	Statistic	df	p-value
X-squared	13.391	8	$p = 0.0991$

```
# =====
# 4. ROC Curve + AUC
# =====
roc_obj <- roc(df_model$LongStay, df_model$pred_prob)
```

```
## Setting levels: control = 0, case = 1
## Setting direction: controls < cases

plot(
  roc_obj,
  col = "black",
  lwd = 2,
  main = "ROC Curve",
  xlab = "1 - Specificity",
  ylab = "Sensitivity"
)

text(
  x = 0.7, y = 0.2,
  labels = paste0("AUC = ", round(auc(roc_obj), 3)),
  cex = 1
)
```



```
billboard$std_resid <- rstandard(m13.6, type = "pearson")
# Count residuals above common thresholds
n_over2 <- sum(abs(billboard$std_resid) > 2)
n_over3 <- sum(abs(billboard$std_resid) > 3)
```

```

max_resid <- max(abs(billboard$std_resid))

# Print results
cat("Residuals > 2:", n_over2, "\n")

## Residuals > 2: 58

cat("Residuals > 3:", n_over3, "\n")

## Residuals > 3: 11

cat("Maximum absolute standardized residual:", round(max_resid, 3), "\n")

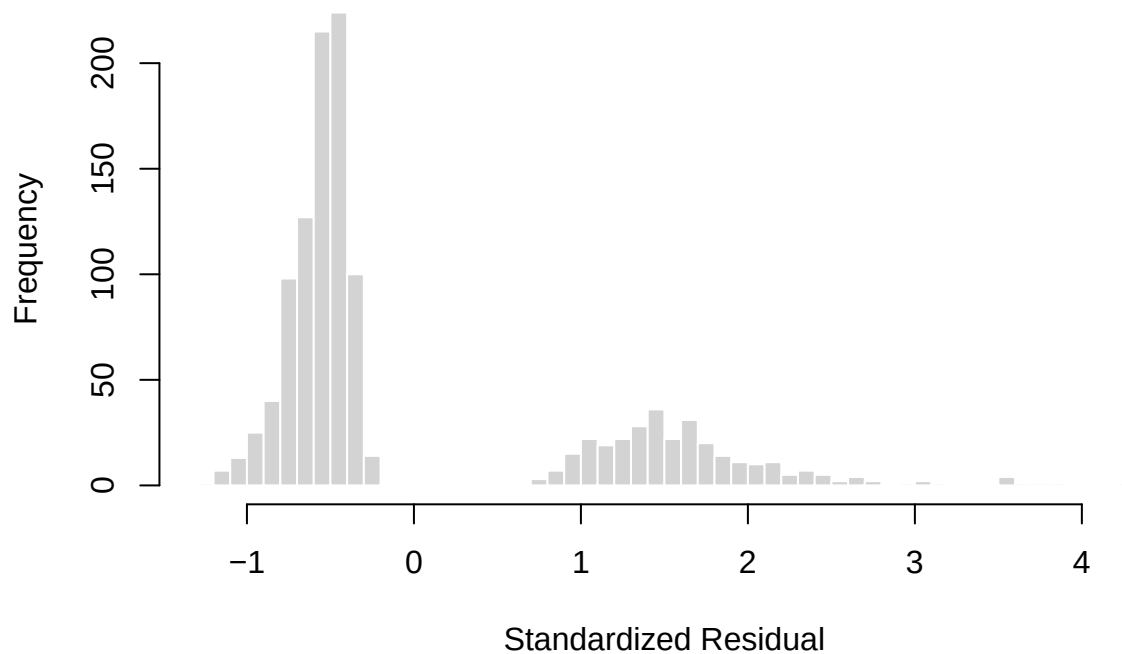
## Maximum absolute standardized residual: 4.248

# Standardized residuals
billboard$std_resid <- rstandard(m13.6, type = "pearson")

# Histogram
hist(billboard$std_resid,
     breaks = 40,
     main = "Histogram of Standardized Residuals",
     xlab = "Standardized Residual",
     col = "lightgray",
     border = "white")

```

Histogram of Standardized Residuals



```
m13i = m13.6
```

```

# Comparison table
models <- list(
  m0 = m0,
  m1 = m1,
  m2 = m2,
  m3 = m3,
  m4 = m4,
  m5 = m5,
  m6 = m6,
  m7 = m7,
  m8 = m8,
  m9 = m9,
  m10 = m10,
  m11 = m11,
  m12 = m12,
  m13 = m13,
  m13i = m13i
)

comparison <- data.frame(
  Model      = names(models),
  DF         = sapply(models, df.residual),
  Deviance   = round(sapply(models, deviance), 3),
  AIC        = round(sapply(models, AIC), 3),
  BIC        = round(sapply(models, BIC), 3)
)

# Produce formatted kable table
library(kableExtra)

kable(
  comparison,
  format = "latex",
  booktabs = TRUE,
  caption = "Comparison of Logistic Regression Models",
  align = c("c", "c", "c", "c", "c")
) %>%
kable_styling(
  latex_options = c("hold_position"),
  position = "center"
)

```

Table 7: Comparison of Logistic Regression Models

	Model	DF	Deviance	AIC	BIC
m0	m0	1151	1274.531	1316.531	1422.927
m1	m1	1152	1275.576	1315.576	1416.906
m2	m2	1153	1275.576	1313.576	1409.839
m3	m3	1154	1275.578	1311.578	1402.774
m4	m4	1155	1277.129	1311.129	1397.259
m5	m5	1156	1277.211	1309.211	1390.275
m6	m6	1157	1277.317	1307.317	1383.314
m7	m7	1158	1277.420	1305.420	1376.350
m8	m8	1159	1277.589	1303.589	1369.453
m9	m9	1160	1277.773	1301.773	1362.570
m10	m10	1161	1278.069	1300.069	1355.800
m11	m11	1162	1278.722	1298.722	1349.387
m12	m12	1163	1280.040	1298.040	1343.638
m13	m13	1164	1281.585	1297.585	1338.117
m13i	m13i	1162	1271.824	1291.824	1342.489